Experiment No: Group A-1

Problem Definition:

Write a program for Tracking Emails and Investigating Email Crimes. i.e. Write a program to analyze e–mail header

1.1 Prerequisite:

Application Layer Protocols

1.2 Learning Objective:

1. To understand how Mails are transferred from Sender to Receiver.
2. To Understand Email related Parameter.

1.3 Theory:

1.3.1 Introduction

Analysis of email is especially important not just because email may be used to communicate about things that we might be interested in for an investigation, but because it is a comparatively permanent and public record of those communications. In the case of a phone call, there is only the record that a call took place; in a spoken conversation, there may be no record at all. Conventional mail can be virtually untraceable, and paper documents are easily destroyed. Email, however, is unique; when a message is sent, the entire message is stored for both the sender and the receiver, and records of the mail being sent are stored on dozens of servers that the message passes through before arriving at its destination. There are a number of ways to analyze email, including: data mining techniques, which may be applied to large or small data sets; straightforward searching of a user's email for certain content; and in-depth analysis of an individual email's lineage.

E-mail system comprises of various hardware and software components that include sender's client and server computers and receiver's client and server computers with required software and services installed on each. Besides these, it uses various systems and services of the Internet. The sending and receiving servers are always connected to the

Internet but the sender's and receiver's client connects to the Internet as and when required. An e-mail communication between a sender '*Alice'* having e-mail address '*alice@a.com*' and recipient '*Bob'* having e-mail address '*bob@b.com*' is shown in figure 1.'*Alice'* composes an e-mail message on her computer called client for '*Bob'* and sends it to her sending server 'smtp.a.org' using *SMTP* protocol. Sending server performs a lookup for the mail exchange record of receiving server '*b.org*' through Domain Name System (*DNS)* protocol on *DNS* server '*dns.b.org*'. The *DNS* server responds with the highest priority mail exchange server '*mx.b.org'* for the domain 'b.org'. Sending server establishes *SMTP* connection with the receiving server and delivers the e-mail message to the mailbox of '*Bob'* on the receiving server. '*Bob'* downloads the message from his mailbox on receiving server to local mailbox on his client computer using *POP3* or *IMAP* protocols. Optionally, '*Bob'* can also read the message stored in his server mailbox without downloading it to the local mailbox by using a Webmail program.

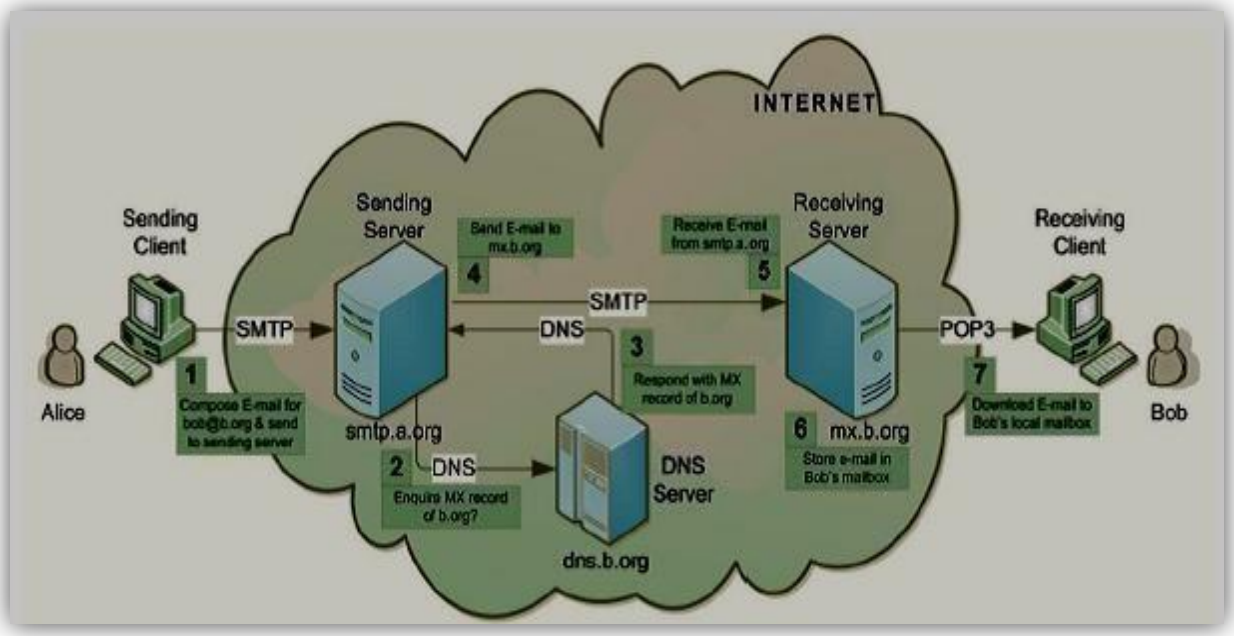


Figure 1*: E-mail communication between a sender 'Alice' and recipient 'Bob'*

1.3.2 E-MAIL ACTORS, ROLES AND RESPONSIBILITIES

E-mail is a highly distributed service involving several actors that play different roles to accomplish end-to-end mail exchange. These actors fall under "User Actors", "Message Handling Service (*MHS*) Actors" and "Administrative Management Domain (*ADMD*) Actors" groups.

User Actors are people, organizations or processes that serve as sources or sinks of messages. They can generate, modify or look at the whole message. User Actors can be of following four types (Table 1):

User Actor Type	Roles and Responsibilities
Author	▪ Responsible for creating the message, its contents, and its list of Recipient addresses. ▪ The MHS transfers the message from the Author and delivers it to the Recipients. ▪ The MHS has an Originator role that correlates with the Author role.
Recipient	▪ The Recipient is a consumer of the delivered message. ▪ The MHS has a Receiver role that correlates with the Recipient role. ▪ A Recipient can close the user-communication loop by creating and submitting a new message that replies to the Author e.g. an automated form of reply is the Message Disposition Notification (MDN)
Return Handler	▪ It is a special form of Recipient that provides notifications (failures or completions) generated by the MHS as it transfers or delivers the message. ▪ It is also called Bounce Handler.
Mediator	▪ It receives, aggregates, reformulates, and redistributes messages among Authors and Recipients. ▪ It forwards a message through a re-posting process. ▪ It shares some functionality with basic MTA relaying, but has greater flexibility in both addressing and content than is available to MTAs. It preserves the integrity and tone of the original message, including the essential aspects of its origination information. It might also add commentary. ▪ It does not create new message that forwards an existing message, Reply or annotation. ▪ Some examples of mediators are: Alias, ReSender, Mailing Lists, Gateways and Boundary Filter.

All types of Mediator user actors set HELO/EHLO, ENVID, RcptTo and Received fields. Alias actors also typically change To/CC/BCC and MailFrom fields. Identities relevant to ReSender are: From, Reply-To, Sender, To/CC/BCC, Resent-From, Resent-Sender, Resent-To/CC/BCC and MailFrom fields. Identities relevant to Mailing List processor are: List-Id, List-*, From, Reply-To, Sender, To/CC and MailFrom fields. Identities relevant to Gateways are: From, Reply-To, Sender, To/CC/BCC and MailFrom fileds.

Message Handling Service (MHS) Actors are responsible for end-to-end transfer of messages. These Actors can generate, modify or look at only transfer data in the message. *MHS* Actors can be of following four types (Table 2):

MHS Actor Type	Roles and Responsibilities
Originator	▪ It ensures that a message is valid for posting and then submits it to a Relay ▪ It is responsible for the functions of the Mail Submission Agent. ▪ It also performs any post-submission that pertain to sending error and delivery notice. ▪ The Author creates the message, but the Originator handles any transmission issues with it
Relay	▪ It performs MHS-level transfer-service routing and store-and-forward function by transmitting or retransmitting the message to its Recipients. ▪ It adds trace information but does not modify the envelope information or the semantics of message content. ▪ It can modify message content representation, such as changing the form of transfer encoding from binary to text, but only (as required) to meet the capabilities of the next hop in the MHS. ▪ When a Relay stops attempting to transfer a message, it becomes an Author because it sends an error message to the Return Address.
Gateway	▪ It connects heterogeneous mail services despite differences in their syntax and semantics. ▪ It can send a useful message to a Recipient on the other side, without requiring changes to any components in the Author's or Recipient's mail services.
Receiver	▪ It performs final delivery or sends the message to an alternate address. ▪ It can also perform filtering and other policy enforcement immediately before or after delivery.

For networks, a port means an endpoint to a logical connection. The port number identifies what type (application/service offered) of port it is. The commonly used default port numbers used in e-mail are shown in Table 3. A complete list of default port numbering assignment is given in

Port No	Protocols/Services	Description
25	SMTP SMTP e-mail server	Simple Mail Transfer Protocol - core Internet protocol used to transfer from client to server (MUA to MTA) and server to server (MTA to MTA)
110	POP3 POP e-mail server	Post Office Protocol allows clients (MUA's) to retrieve stored e-mail
143	IMAP IMAP(4) e-mail server	Internet Message Access Protocol provides a means of managing e-mail messages on a remote server and retrieve stored e-mail
465	SMTPS WSMTP (SSMTP) protocol over TLS|SSL	SMTP via SSL encrypted connection (Unofficial)
993	IMAPS SSL encrypted IMAP	IMAP via SSL encrypted connection
995	POP3S SPOP SSL encrypted POP	POP via SSL encrypted connection
587	MSA	Outgoing Mail (Submission)
80	HTTP	Webmail
443	HTTPS	Secure Webmail

1.3.3 Analyzing an Individual Email

Although webmail will feature prominently in this section, the analysis of a particular email's lineage is much broader and can be applied to any email. A simple view of the path of an email from a sender to a client is presented in Figure 2. The email originates from the sender, whether from a local email client or a webmail application. When the email is sent, it is first sent to a Simple Mail Transfer Protocol (SMTP) server. That server forwards it to other SMTP servers until it finally reaches the destination server. On reaching its destination, the email is sent to a Post Office Protocol (POP) server, or any number of similar mail-delivery servers (IMAP is another, and webmail services may use their own servers for this purpose). The receiving client then connects to that server, retrieves the message, and allows the recipient to read it.

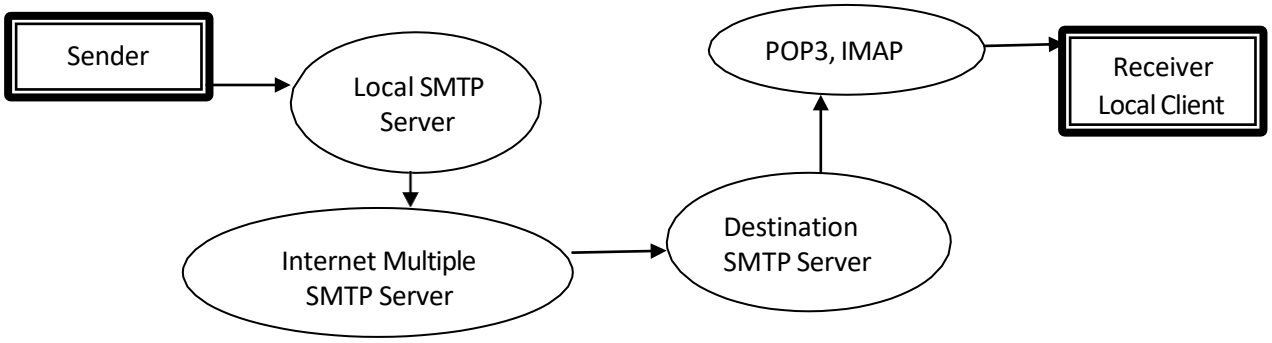


Figure 2: *Path of an email from a sender to a client*

When the email is sent and when it is received, those respective servers add their own information to the email's header, and most likely log the action. Access to those logs may be required for much analysis, but specifics are outside of the scope of this paper. Considerable information can be gleaned from the header alone.

Suppose Moses, with the address moses@nmt.edu, sends an email from his office on the New Mexico Tech campus to his similarly named friend, with the email address thenewmoses@gmail.com. The subject of this email is "Snakes," and the content "Fish."

Below is the entire theoretical email, including all headers.

From:
moses@nmt.edu
Subject: Snakes
Date: September 25, 2021 9:35:29 PM
MDT To: thenewmoses@gmail.com
X-Gmail-Received:
ca493ed685a8e9ae77165ab2ce345127e5b310b4 Delivered-
To: thenewmoses@gmail.com
Received: by 10.90.33.15 with SMTP id g15cs279684agg; Mon, 25 Sep
2021 20:35:32 0700 (PDT)
Received: by 10.35.113.12 with SMTP id q12mr526602pym; Mon, 25 Sep 2021
20:35:32 -0700 (PDT)
Received: from mailhost.nmt.edu (mailhost.NMT.EDU
[129.138.4.52]) by mx.gmail.com with ESMTP id 36si2059018nza.
2021.09.25.20.35.32; Mon, 25 Sep 2021 20:35:32 -0700 (PDT)
Received: from localhost (localhost.localdomain [127.0.0.1]) by
localhost.localdomain (Postfix) with ESMTP id 09FF4436164 for
<thenewmoses@gmail.com>; Mon, 25 Sep 2021 21:35:32 -0600
(MDT) Received: from mailhost.nmt.edu ([127.0.0.1]) by localhost
(mailhost.nmt.edu [127.0.0.1]) (amavisd-new, port 10024) with
ESMTP id 11225-05 for <thenewmoses@gmail.com>; Mon, 25 Sep 2021
21:35:30 -0600 (MDT)
Received: from [192.168.1.2] (cs-fitch017.nmt.edu [129.138.21.110])
by mailhost.nmt.edu (Postfix) with ESMTP id 6FD4B436030 for
<thenewmoses@gmail.com>; Mon, 25 Sep 2021 21:35:30 -0600
(MDT) Return-Path: <moses@nmt.edu>
Received-Spf: pass (gmail.com: best guess record for
domain of moses@nmt.edu designates 129.138.4.52 as
permitted sender) Mime-Version: 1.0 (Apple Message
framework v752.2)
Content-Transfer-Encoding: 7bit
Message-Id: <77E313EF-271F-4AD0-A8D3-81263BF7B083@nmt.edu>

Content-Type: text/plain; charset=US-ASCII;
format=flowed X-Mailer: Apple Mail (2.752.2)
X-Virus-Scanned: by amavisd-new-2.3.1 (20050509) (RHEL AS) at
nmt.edu Fish

E-MAIL IDENTITIES:

Field Name	Set By	Field Description
Layer: ***Message Header Fields (Identification Fields)***		
Message- ID:	Originator	Globally unique message identification string generated when it is sent.
In-Reply-To:	Originator	Contains the Message-ID of the original message in response to which the reply message is sent.
References:	Originator	Identifies other documents related to this message, such as other e-mail message.
Layer: ***Message Header Fields (Originator Fields)***		
From:	Author	Name and e-mail address of the author of the message
Sender:	Originator	Contains the address responsible for sending the message on behalf of Author, if not omitted or same as that specified in From field.
Reply- To:	Author	E-mail address, the author would like recipients to use for replies. If present it overrides the From field.
Layer: ***Message Header Fields (Originator Date Fields)***		
Date:	Originator	It holds date and time when the message was made available for delivery.
Layer: ***Message Header Fields (Informational Fields)***		
Subject:	Author	It describes the subject or topic of the message.
Comments:	Author	It contains summarized comments regarding the message.
Keyword:	Author	It contains list of comma separated keywords that may be useful to the recipients e.g. when searching mail.
Layer: ***Message Header Fields (Destination Address Fields)***		
TO:	Author	Specifies a list of addresses of the recipients of the message. These addresses might be different from address in RcptTo SMTP commands
CC:	Author	Generally same as To Field. Generally a To field specifies primary recipient who is expected to take some action and CC addresses

1.4 Execution Steps

- Open the Email which you want to analyze header
- Click on the right side three vertical dot(more) and select show original.
- New tab will be open then copy header information which you want to analyze.
- Open **https://www.whatismyip.com/email-header-analyzer** website.
- Copy the header information and click on analyze button.
- Then You will see the header analysis on screen.

Assignment Group A-2

Problem Definition:

Implement a program to generate and verify CAPTCHA image.

1.1 Prerequisite:

Basics of PYTHON

1.2 Learning Objectives:

1. Understand the use of CAPTCHA Image.
2. Generation and Verification of it.

1.3 New Concepts:

1. CAPTCHA generation
2. Functions used like RANDOM

1.4 Theory

1.4.1 Introduction

A **CAPTCHA** (an acronym for "**C**ompletely **A**utomated **P**ublic **T**uring test to tell **C**omputers and **H**umans **A**part") is a type of challenge-response test used in computing to determine whether or not the user is human. The most common type of CAPTCHA was first invented by Mark D. Lillibridge, Martin Abadi, Krishna Bharat and Andrei Z. Broder. This form of CAPTCHA requires that the user type the letters of a distorted image, sometimes with the addition of an obscured sequence of letters or digits that appears on the screen. Because the test is administered by a computer, in contrast to the standard Turing test that is administered by a human, a CAPTCHA is sometimes described as a reverse Turing test. Actually CAPTCHA is used as a simple puzzle hurdle, which restricts various automated

programs to sign-up E-mail accounts, cracking passwords, spam sending, privacy violation etc. This CAPTCHA actually challenges a particular automated program, which is trying to access some private zone. So, CAPTCHA helps in preventing access of personal mail accounts by some un-authorized automated spamming programs

1.4.2 Characteristics:-

CAPTCHAs are by definition fully automated, requiring little human maintenance or intervention to administer. This has obvious benefits in cost and reliability.

By definition, the algorithm used to create the CAPTCHA must be made public, though it may be covered by a patent. This is done to demonstrate that breaking it requires the solution to a difficult problem in the field of artificial intelligence (AI) rather than just the discovery of the (secret) algorithm, which could be obtained through reverse engineering or other means.

Modern text-based CAPTCHAS are designed such that they require the simultaneous use of three separate abilities—invariant recognition, segmentation, and parsing—to correctly complete the task with any consistency.

1. Invariant recognition refers to the ability to recognize the large amount of variation in the shapes of letters. There are nearly an infinite number of versions for each character that a human brain can successfully identify. The same is not true for a computer, and teaching it to recognize all those differing formations is an extremely challenging task.
2. Segmentation, or the ability to separate one letter from another, is also made difficult in CAPTCHAs, as characters are crowded together with no white space in between.
3. Context is also critical. The CAPTCHA must be understood holistically to correctly identify each character. For example, in one segment of a CAPTCHA, a letter might look like an “m.” Only when the whole word is taken into context does it become clear that it is a “u” and an “n.”

Each of these problems pose a significant challenge for a computer, even in isolation. The presence of all three at the same time is what makes CAPTCHAs difficult to solve.

1.4.3 Why we Prefer Captcha rather other security measures?

1. To Protect Website's Registration Forms –

Many Websites like Hotmail, Gmail, Yahoo, Facebook etc. offers free free registration. It is necessary to protect these website's registrations so that it ensures the registered user is a human not a program or bot. Captcha Code is used to protect the Registration Form Submission Programmatically.

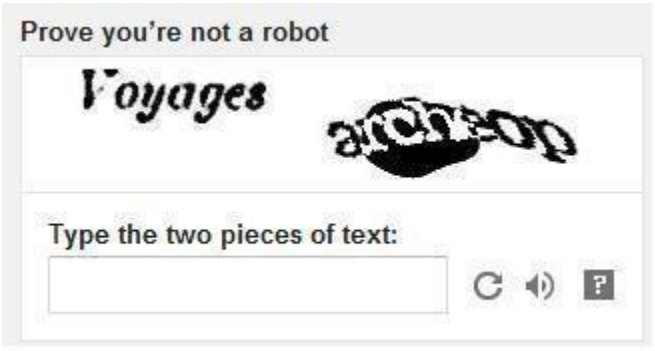


Figure 1:- *Gmail Registration Form Captcha Code Image Screen Capture*

2. To Prevent Comment Spams in Blogs –

Captcha Code is used in the comment form so that only human can comment on a post otherwise spammers can flood hundreds of comments to a single post.

3. To Protect Email Address Scrapping –

Spammers crawl the web in the search of Email address posted in the clear text (e.g. **email@website.com**). You can protect your email address either by using Captcha to hide the email address, one can solve the Captcha before showing the Email address or by using alternative trick to post Email Address in the format of email at website dot com.

4. To Protect from Search Engine Bots –

Many Html tags are available to specifying indexing condition to Search engine bots. To prevent a website or specific webpage from search engine crawling, it is desirable to use

html meta tag but sometimes it is not completely sure that the webpage is fully protected from search engine crawlers and large companies who needs a high security uses Captcha rather than to use only meta tags to protect their highly public protected and confidential Web Pages.

1.4.4 Application of CAPTCHA:-

Applications of CAPTCHAs

CAPTCHAs have several applications for practical security:

- **Preventing Comment Spam in Blogs.** Most bloggers are familiar with programs that submit bogus comments, usually for the purpose of raising search engine ranks

  of some website (e.g., "buy penny stocks here"). This is called comment spam. By using a CAPTCHA, only humans can enter comments on a blog. There is no need to make users sign up before they enter a comment, and no legitimate comments are ever lost!
- **Protecting Website Registration.** Several companies (Yahoo!, Microsoft, etc.) offer free email services. Up until a few years ago, most of these services suffered from a specific type of attack: "bots" that would sign up for thousands of email accounts every minute. The solution to this problem was to use CAPTCHAs to ensure that only humans obtain free accounts. In general, free services should be protected with a CAPTCHA in order to prevent abuse by automated scripts.
- **Protecting Email Addresses From Scrapers.** Spammers crawl the Web in search of email addresses posted in clear text. CAPTCHAs provide an effective mechanism to hide your email address from Web scrapers. The idea is to require users to solve a CAPTCHA before showing your email address.
- **Online Polls.** In November 1999, *http://www.slashdot.org* released an online poll asking which was the best graduate school in computer science (a dangerous question to ask over the web!). As is the case with most online polls, IP addresses of voters were recorded in order to prevent single users from voting more than once. However, students at Carnegie Mellon found a way to stuff the ballots using

programs that voted for CMU thousands of times. CMU's score started growing rapidly. The next day, students at MIT wrote their own program and the poll became a contest between voting "bots." MIT finished with 21,156 votes, Carnegie Mellon with 21,032 and every other school with less than 1,000. Can the result of any online poll be trusted? Not unless the poll ensures that only humans can vote.

- **Preventing Dictionary Attacks.** CAPTCHAs can also be used to prevent dictionary attacks in password systems. The idea is simple: prevent a computer from being able to iterate through the entire space of passwords by requiring it to solve a CAPTCHA after a certain number of unsuccessful logins. This is better than the classic approach of locking an account after a sequence of unsuccessful logins, since doing so allows an attacker to lock accounts at will.
- **Search Engine Bots.** It is sometimes desirable to keep WebPages unindexed to prevent others from finding them easily. There is an html tag to prevent search engine bots from reading web pages. The tag, however, doesn't guarantee that bots won't read a web page; it only serves to say "no bots, please." Search engine bots, since they usually belong to large companies, respect web pages that don't want to allow them in. However, in order to truly guarantee that bots won't enter a web site, CAPTCHAs are needed.
- **Worms and Spam.** CAPTCHAs also offer a plausible solution against email worms and spam: "I will only accept an email if I know there is a human behind the other computer." A few companies are already marketing this idea.

1.4.5 Advantages:

1. Distinguishes between a human and a machine
2. Makes online polls more legitimate
3. Reduces spam and viruses
4. Makes online shopping safer
5. Diminishes abuse of free email account services

1.4.6 Disadvantages:

1. Sometimes very difficult to read
2. Are not compatible with users with disabilities
3. Time-consuming to decipher
4. Technical difficulties with certain internet browsers
5. May greatly enhance Artificial Intelligence

1.5 Algorithm:

1. Start
2. Import **image.captcha** from ImageCaptcha
3. Generate captcha by **generate(captcha_text)**
4. For randomly captcha generation use function **random.randint()**
5. Create an image to write a captcha text
6. End

Assignment Group A-3

Problem Definition: Write a Computer Forensics Application Program in Java/Python/C++ for recovering Deleted Files and Deleted Partitions.

3.1 Prerequisite:

a) Knowledge about Partitions in Ubuntu. b) Path of Trash folder.

3.2 Learning Objectives:

- Understand the concept of Recovery of deleted files.
- Implementation of recovery of deleted Partitions.

3.3 New Concepts:

a. Recovery of files in LINUX OS.

3.4 Theory

Introduction

Have you accidentally deleted an important file because you are in a habit of using "Shift+Del" rather than delete only?? Well don't panic.There are many utilities in Ubuntu and other Linux distributions which helps you in recovering the so called "permanently deleted" files. Actually when you delete a file permanently (accidentally or intentionally), It doesn't get removed from your hard disk.It get stored in certain blocks of the storage device and they continue to exist in the blocks unless you overwrite them with newer files. There are many Tools available to recover permanently deleted files Scalpel.

Scalpel is a platform independent command based tool which is small yet very powerful. But, if the file is deleted i.e. by just pressing Delete button the file is stored in Trash folder in Ubuntu OS. So it is easy to recover the deleted files from Trash Folder. Just we need to know the path of trash folder.

Path is: ="/home/gurukul/.local/share/Trash/files"

There are sub-Folders in Trash Folder namely :

1. files- contains files which are deleted
2. info- contains information of files deleted
3. expunged

Introduction to file systems:

File systems are one of the things any newcomer to linux must become acquainted with. In the world of Microsoft you never really have to worry about it, the default being NTFS. Linux however, being built on a world of open source and differing opinions, is not limited in this way and so the user should have an understanding of what a file system is, and how it affects the computer.

At the core of a computer, it's all 1s and 0s, but the organization of that data is not quite as simple. A *bit* is a 1 or a 0, a *byte* is composed of 8 bits, a kilobyte is 1024 (i.e. 2) bytes, a megabyte is 1024 kilobytes and so on and so forth. All these *bits* and *bytes* are permanently stored on a Hard Drive. A hard drive stores all your data, any time you save a file, you're writing thousands of 1s and 0s to a metallic disc, changing the magnetic properties that can later be read as 1 or 0. There is so much data on a hard drive that there has to be some way to organize it, like a library of books and the old card drawers that indexed all of them, without that index, we'd be lost. Libraries, for the most part, use the Dewey Decimal System to organize their books, but

there exist other systems to do so, none of which have attained the same fame as Mr. Dewey's invention. File systems are the same way. The ones most users are aware of are the ones Windows uses, the vFat or the NTFS systems, these are the Windows default file systems.

Ubuntu (like all UNIX-like systems) organizes files in a hierarchical tree, where relationships are thought of in teams of children and parent. *Directories* can contain other directories as well as *regular files*, which are the "leaves" of the tree. Any element of the tree can be references by a *path name*; an *absolute path name* starts with the character / (identifying the *root directory*, which contains all other directories and files), then every child directory that must be traversed to reach the element is listed, each separated by a / sign.

Main directories

The standard Ubuntu directory structure mostly follows the File system Hierarchy Standard, which can be referred to for more detailed information.

Here, only the most important directories in the system will be presented.
/bin is a place for most commonly used terminal commands, like ls, mount, rm, etc.

/boot contains files needed to start up the system, including the Linux kernel, a RAM disk image and bootloader configuration files.

/dev contains all *device files*, which are not regular files but instead refer to various hardware devices on the system, including hard drives.

/etc contains system-global configuration files, which affect the system's behavior for all users. **/home** home sweet home, this is the place for users' home directories.

/lib contains very important dynamic libraries and kernel modules

/media is intended as a mount point for external devices, such as hard drives or removable media (floppies, CDs, DVDs).

/mnt is also a place for mount points, but dedicated specifically to "temporarily mounted" devices, such as network filesystems.

/opt can be used to store addition software for your system, which is not handled by the package manager.

/proc is a virtual filesystem that provides a mechanism for kernel to send information to processes.

/root is the superuser's home directory, not in /home/ to allow for booting the system even if /home/ is not available.

/sbin contains important administrative commands that should generally only be employed by the superuser.

/srv can contain data directories of services such as HTTP (/srv/www/) or FTP.

/sys is a virtual filesystem that can be accessed to set or obtain information about the kernel's view of the system.

/tmp is a place for temporary files used by applications.

/usr contains the majority of user utilities and applications, and partly replicates the root

directory structure, containing for instance, among others, /usr/bin/ and /usr/lib.

/var is dedicated variable data that potentially changes rapidly; a notable directory it contains is /var/log where system log files are kept. **Steps to Partition HardDisk Drive in Ubuntu:-**

Step 1. If you are trying to format or partition your hard drive it is assumed that bios is able to detect the device. To determine the path and other specific information about your drive open a terminal window and enter this command:

sudo lshw -C disk

Step 2. After entering this command Ubuntu should return something similar to this. Take note of the "logical name" because this will be used throughout the partitioning process if done via terminal window**.**



Step 3. The part we will be most concerned with will be the hard drive information that is displayed in the terminal window.



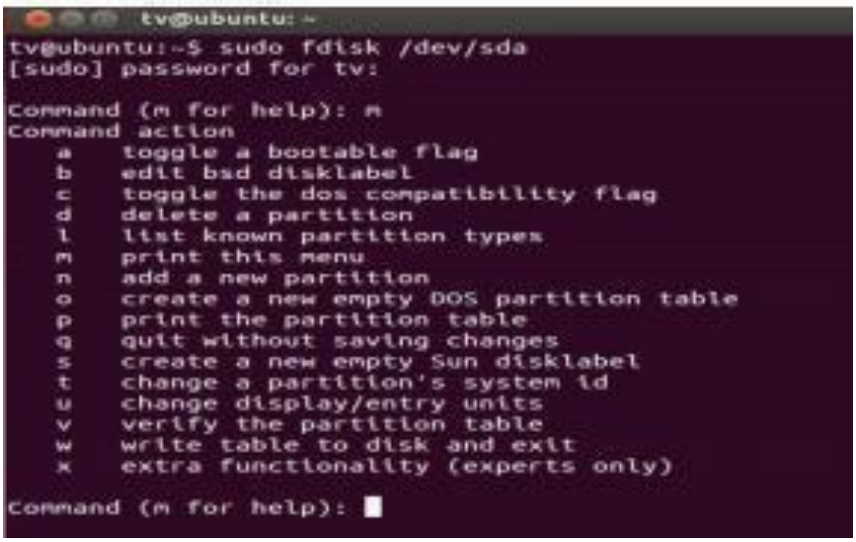
If you plan on using the hard drive only for Ubuntu then the recommended file system to use is either ext3/ext4 depending on whether or not you need backwards compatibility with previous versions of Linux. If you will need to share files between Ubuntu and Windows machines fat 32 is the recommended file system to use, but NTFS will also work well also.

Partition using command line in Terminal: Step

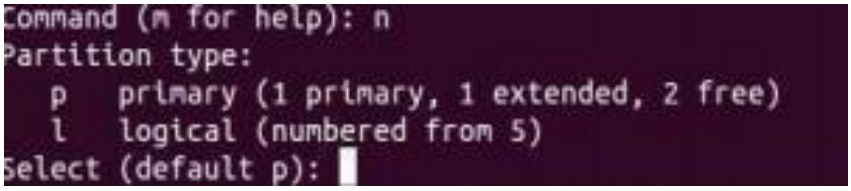
1. Start fdisk with this command



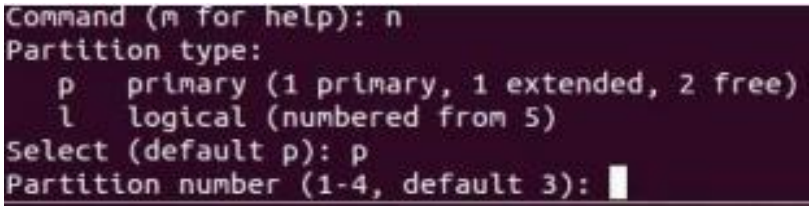
Step 2. Press **"m"** then hit **enter**. This will return a menu like the one below showing all of the available commands for the fdisk program.



Step 3. Since we want to add a new partition press **"n"** and then **enter**.



Step 4. To create a primary partition (what we want) press **"p"** and then hit **enter**.



Step 5.

If you only want 1 partition press **"1"** and hit **enter**. You may be provided with a default response, you may choose this as the Partition number if you would like. Next you will be prompted for the locations of where you would like the first and last sectors of the partition to be. You may again be provided with default responses choose these if you want.

Step 6.

Now choose w to write the partition to the disk. Type "w" then press enter.Your drive is now partitioned. Now we need to format it. By default Linux will recognize this partition as dev/sdb1.

Step 7. To format the partition with an ext3 filesystem.

sudo mkfs -t ext3 /dev/sdb1

3.5 Algorithm:

1. Start
2. Initialize variables as path="/home/gurukul/.local/share/Trash/files"
   infopath="/home/gurukul/.local/share/Trash/info"
3. Check the list of files present in files folder.
4. Find the path of file to restore it using info folder.
5. Copy the contents of file which is deleted and is in Trash folder into new file at original location.
6. Delete the file from Trash folder.
7. End

3.6 Mathematical Model:

I= P (path of trash folder) **Functions:**
re.findall(r'/.*',line)
destipath.lstrip('[') destipath.rstrip(']')
destipath[:-1]
destipath[1:] **Output:**
Recovered file

Assignment Group A-4

Write a program for Log Capturing and Event Correlation

Prerequisite:

- Latest version of Squid should be used.(version 2.5 or greater)
- A web server for testing purpose which can be used instead of Internet.
- Squid Version greater than 2.6 is required for Transparent squid proxy configuration in this lab.
-

Learning Objectives:

- To understand how Log Records are generated for Further Analysis.

New Concepts:

- Squid and Sarg

Theory

1.1. Introduction:

- During the period of development of internet, users are allowed for unlimited access to the resources due to less number of users. So there were less issues related to accessing speed over internet.
- With the increase in internet usage, many issues raised related to accessing speed, effective bandwidth utilization etc. One method of overcoming these issues is, maintaining a copy of webpage visited by a user in the cache so that the other user who visits the same webpage will access the same website within a short period of time. This method not only increases the accessing speed but also helps in utilizing the bandwidth effectively.
- The above said functionality can be achieved by maintaining a proxy server through which all the users in the organization or a group access the internet. The most widely used proxy server in Linux is Squid Proxy, which is free software released General Public License.

- Squid provides proxy and cache services for **H**yper **T**ext **T**ransfer **P**rotocol (HTTP),

  File **T**ransfer **P**rotocol (FTP), and various other protocols.

To configure a system as a proxy server, one should have a sufficient amount of memory for maintaining the cache which in turn increases the performance.

- In case if the internet connection is not available, setup one host as a web server in place of internet and assign the IP address to the proxy server network interface in the network, used by web server instead of public IP address assigned to that interface.

1.2. Steps to Configure Squid Proxy:

1.2.1. Installation of Squid Package

A Squid proxy server is generally installed on a separate server than the Web server with the original files. Squid works by tracking object use over the network. Squid will initially act as an intermediary, simply passing the client's request on to the server and saving a copy of the requested object. If the same client or multiple clients request the same object before it expires from Squid's cache, Squid can then immediately serve it, accelerating the download and saving bandwidth.

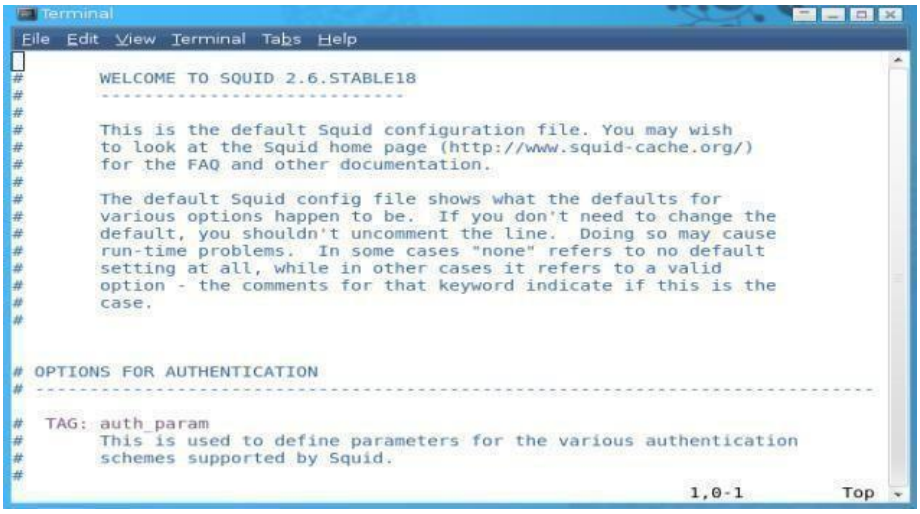
```
sudo apt update
sudo apt -y install squid
```

1.2.2. Accessing the Proxy Server configuration file

To configure squid proxy server we need to edit the *sudo gedit /etc/squid/squid.conf* file and the default location of squid.conf file varies from distribution to distribution and from version to version. We can edit the configuration file using vi editor through command prompt.

```
sudo gedit /etc/squid/squid.conf
```

Then the content of the configuration file can be viewed as shown below in the figure.



Editing the squid configuration file

sudo gedit /etc/squid/squid.conf

Search the TAG: auth_param and paste the following acl

```
auth_param basic program /usr/lib/squid/basic_ncsa_auth /etc/squid/passwd
auth_param basic children 5
auth_param basic realm Squid Basic Authentication
auth_param basic credentialsttl 2 hours
acl auth_users proxy_auth REQUIRED
http_access allow auth_users

 #search localnet and paste line in last

acl localnet src 192.168.60.70
```

>sudo service squid3 restart

Specifying the interface and port number on which the proxy server should listen.

By default, the proxy server will listen on all the available network interfaces on the system for requests. For Example, if one interface card is assigned a public ip from which it is connected to internet and the other interface card is assigned an ip address which belongs to your local area network. Then in order to make you proxy server to listen for requests from your Local Area Network through a particular port, then change the variable http_port 3128 in the squid configuration file to desired ip address and port number in the format shown below.

http_port <ip address belonging to LAN>:<port number>

Example: For example, if your proxy server has an ip address 192.168.60.70 which belongs to the local area network 192.168.60.0/24 and you want the server to listen for requests from your LAN through a particular port say 3456, then you can change the variable http_port as shown.

http_port 192.168.60.70:3456

Assigning Access Controls

By default, no user machine is allowed to connect to the proxy server except the localhost. To allow the local machines access your proxy server, locate the acl section in the squid configuration file starting with acl and at the end of the last acl line specify your access

control. For example to allow local area network 192.168.60.0/24 machines to access your proxy server, specify the acl as

acl mylan src 192.168.60.0/255.255.255.0

In the above example, mylan specifies the name of my access control. We can specify any name other than my lan for access control. src specifies the source network.

Allow or Deny based on Access Control.

After specifying the access control for your local LAN, we need to provide allow permission for the specified LAN using http_access variable in the squid configuration file as shown in the example below.

Example: To allow the above specified access control (i.e acl mylan src 192.168.60.0/255.255.255.0), we need to specify the http_access variable as

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Hyderabad http_access allow mylan

Here mylan specifies the access control used. Suppose if we want to allow all the networks except the 192.168.60.0/24 network to access the proxy then we can specify the http_access variable as

http_access deny !mylan

In the above line, !mylan specifies except mylan network.

Note:

The above specified http_access variable should be specified before the line

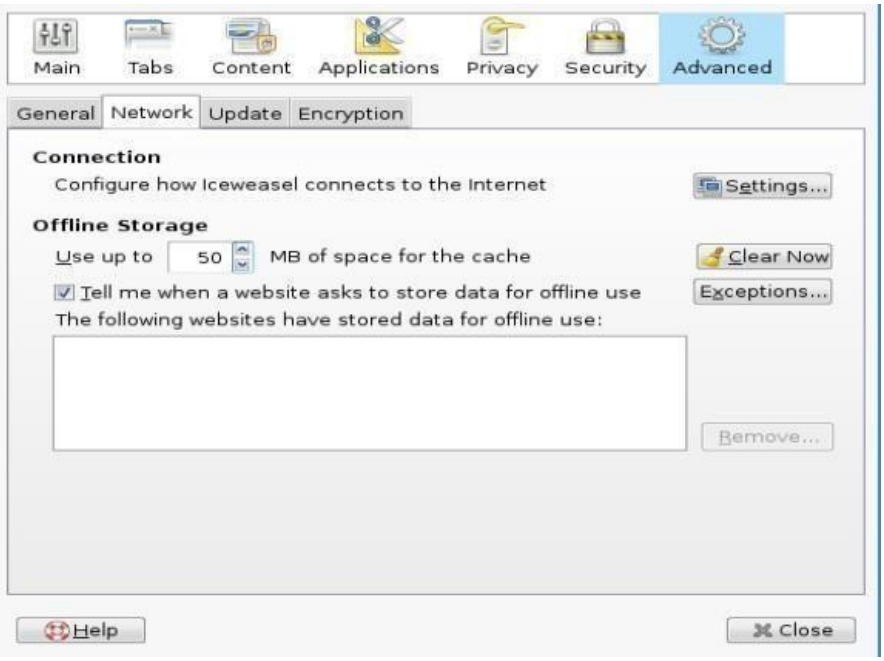
http_access deny all in the configuration file.

Saving the changes and exit the gedit Editor

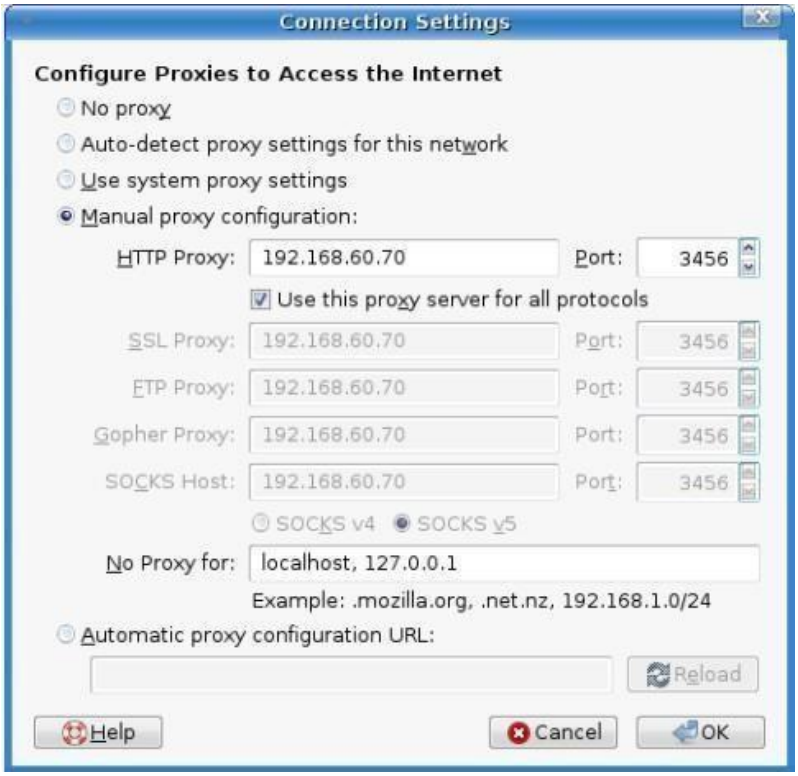
After making appropriate changes to your configuration file exit the vi editor window by pressing Esc followed by :wq!. Here wq specifies save changes and exit the configuration file.

4.3 Testing the Squid configuration

To test the squid configuration, open a browser in any one of the pc in local area network or on the proxy server and specify the proxy settings as the ipaddress of the proxy server and port on which it is listening for requests. For example, in firefox web browser if we want to set the proxy settings in the browser window goto **Edit** -->**Preferences** and window similar to shown below will be displayed.



Now select Advanced tab, and under advanced tab click on Network tab and click on Settings option under Connection field. Then a window similar to the shown below will be displayed.



4.4SARG – Squid Analysis Report Generator and Internet Bandwidth Monitoring Tool

SARG is an open source tool that allows you to analyze the squid log files and generates beautiful reports in HTML format with information's about users, IP addresses, top accessed sites, total bandwidth usage, elapsed time, downloads, access denied websites, daily reports, weekly reports and monthly reports.

The SARG is very handy tool to view how much internet bandwidth is utilized by individual machines on the network and can watch on which websites the network's users are accessing.

Installing Sarg from Source

The '**sarg**'package by default not included in **RedHat** based distributions, so we need to manually compile and install it from source tar ball. For this, we need some additional pre- requisites packages to be installed on the system before compiling it from source.

$ sudo apt-get install sarg

Configuring Sarg

Now it's time to edit some parameters in SARG main configuration file. The file contains lots of options to edit, but we will only edit required parameters like:

Access logs

path

Output

directory

Date

Format

Overwrite report for the same date.

Open sarg.conf file with your choice of editor and make changes as shown below.

vi /usr/local/etc/sarg.conf [On RedHat based systems]

Now Uncomment and add the original path to your squid access
log file. # sarg.conf

TAG: access_log file
Where is the access.log file
sarg
-l file
access_log /var/log/squid/access.log

Next, add the correct Output directory path to save the generate squid reports in that directory. Please note, under Debian based distributions the Apache web root directory is '/var/www'. So, please be careful while adding correct web root paths under your Linux distributions.

TAG: output_dir
The reports will be saved in that
directory # sarg -o dir

output_dir /var/www/html/squid-reports

Set the correct date format for reports. For example, 'date_format e' will

display reports in 'dd/mm/yy' format.

TAG: date_format

Date format in reports: e (European=dd/mm/yy), u (American=mm/dd/yy), w (Weekly=yy.ww)

#

date_format e

Next, uncomment and set Overwrite report to 'Yes'. # TAG: overwrite_report yes|no

yes - if report date already exist then will be overwritten.

no - if report date already exist then will be renamed to filename.n, filename.n+1

overwrite_report yes

That's it! Save and close the file.

Step 3: Generating Sarg Report

Once, you've done with the configuration part, it's time to generate the squid log report using the following command.

sarg -x [On RedHat based systems]

Assessing Sarg Report

The generated reports placed under '/var/www/html/squid-reports/' or '/var/www/squid-reports/' which can be accessed from the web browser using the address.

http://localhost/squid-reports OR

http://ip-address/squid-reports

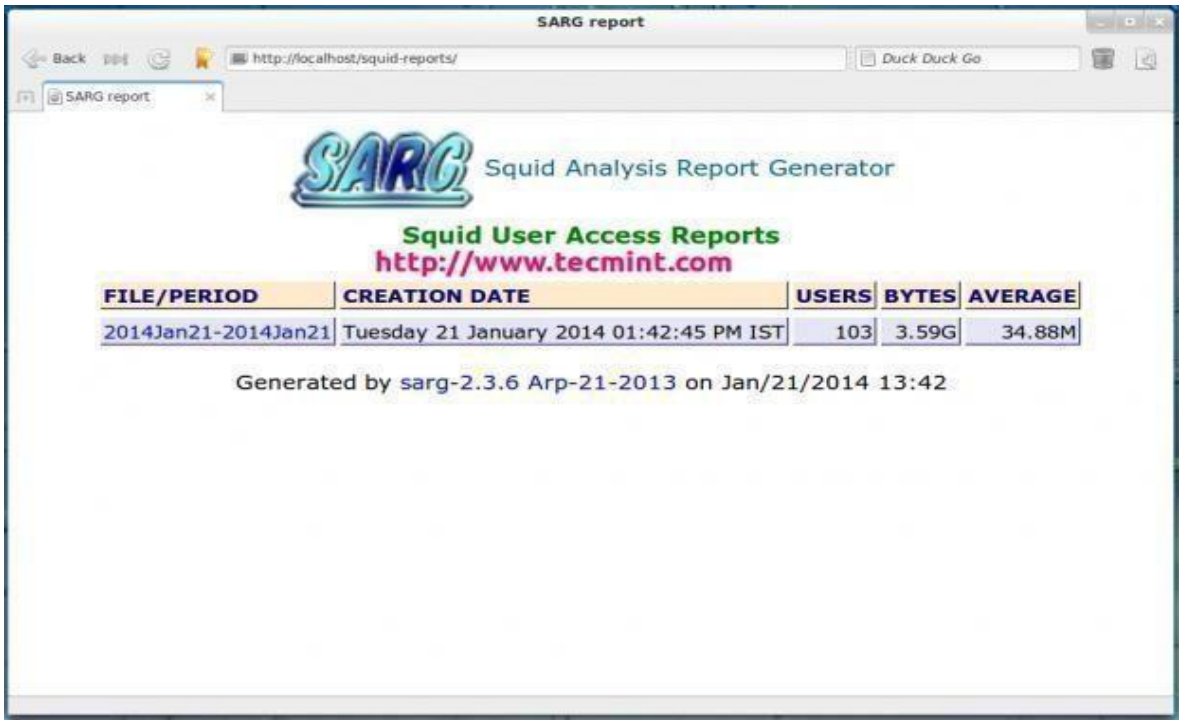


Fig.1 Sarg Main Window

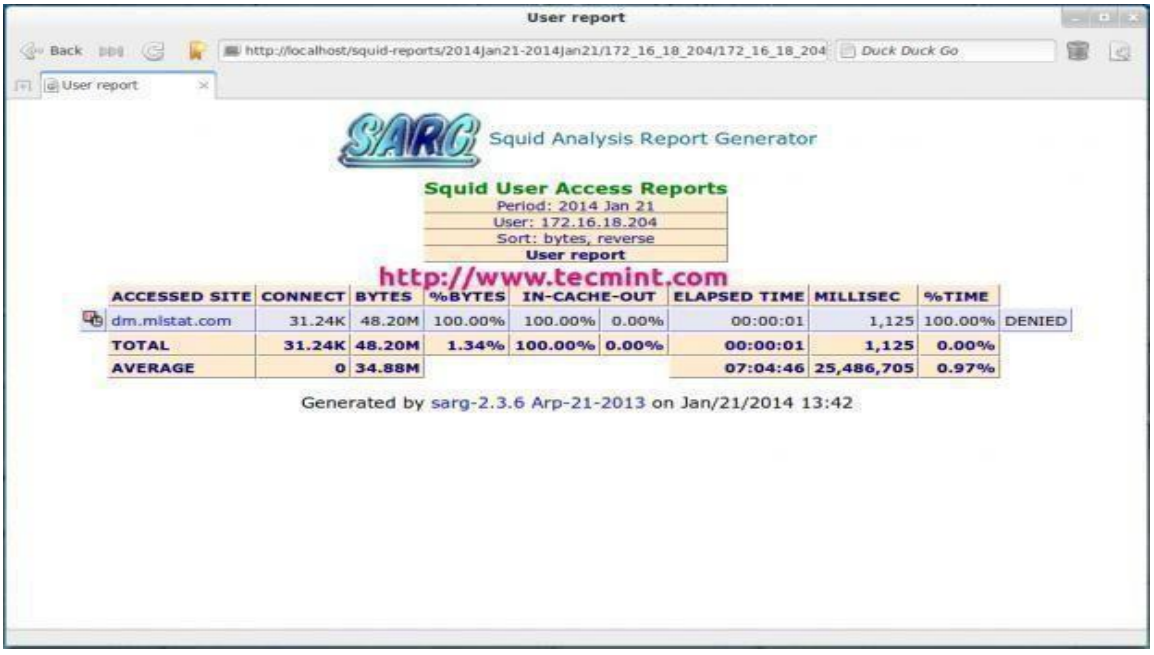
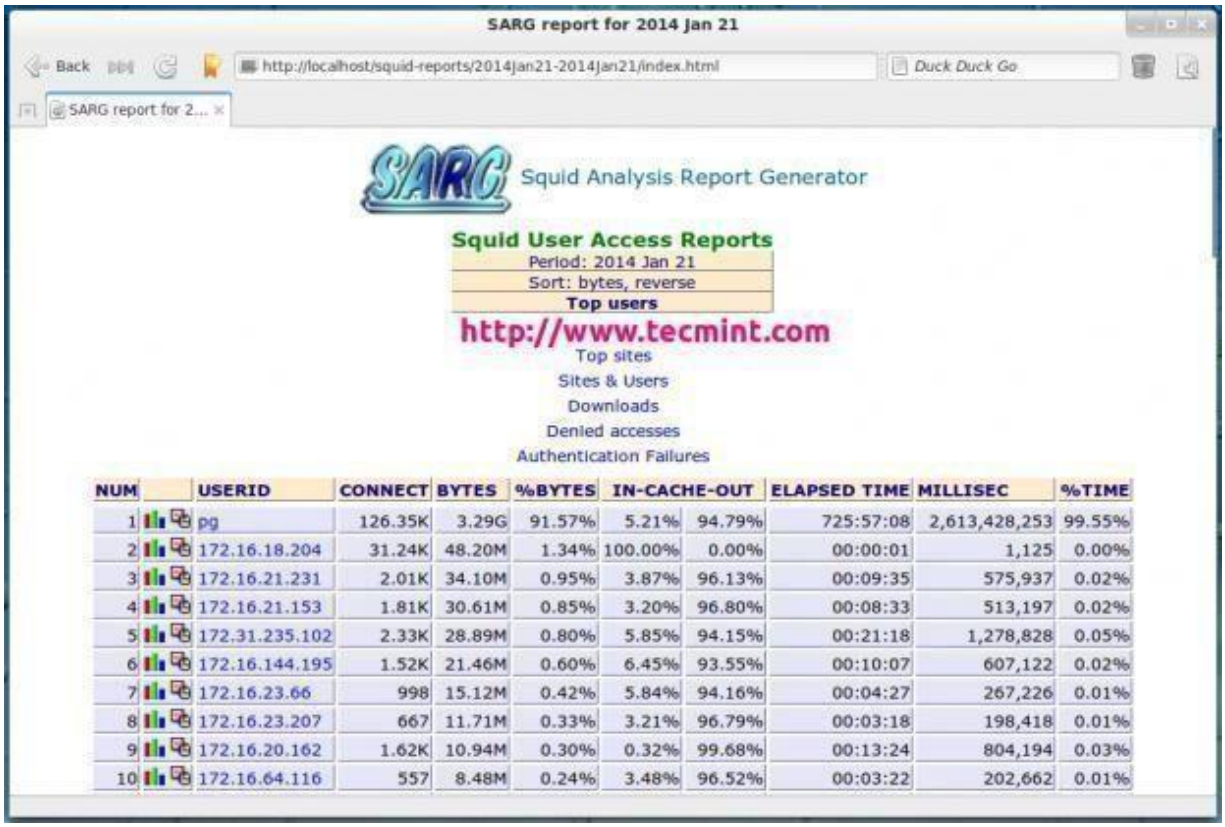
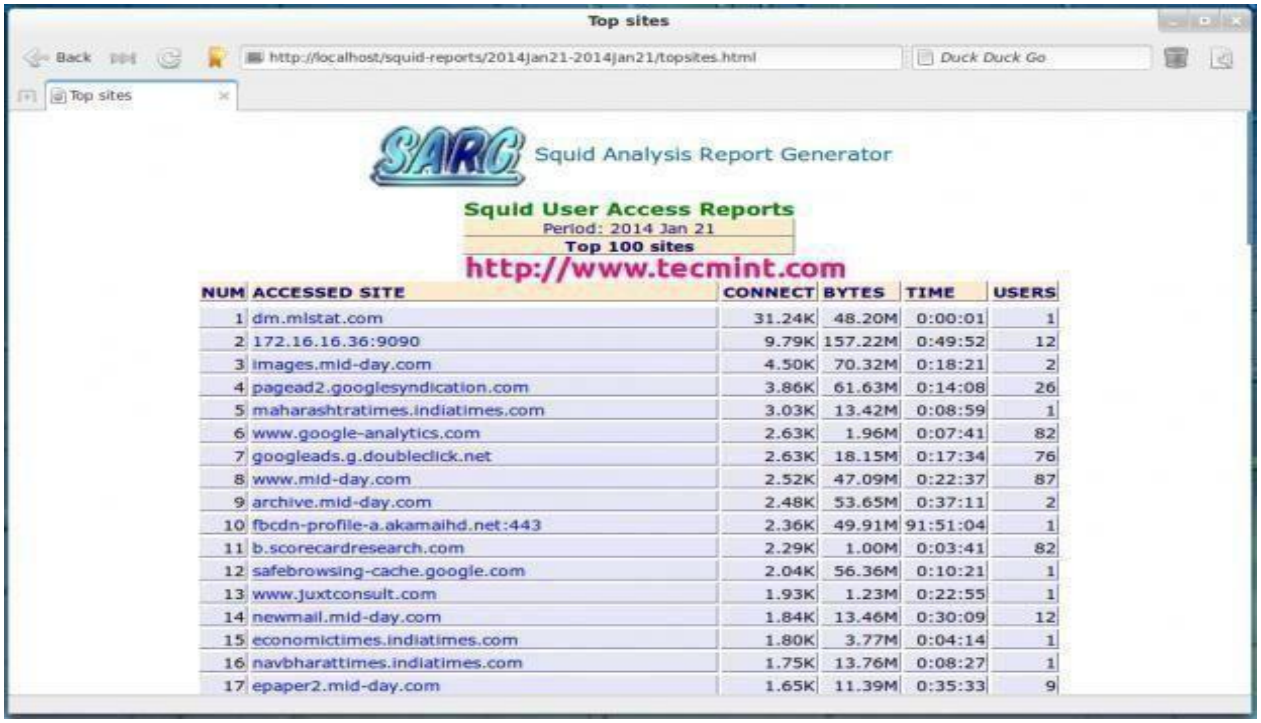


Fig2.User Report



NUM		USERID	CONNECT	BYTES	%BYTES	IN-CACHE-OUT		ELAPSED TIME	MILLISEC	%TIME
1		pg	126.35K	3.29G	91.57%	5.21%	94.79%	725:57:08	2,613,428,253	99.55%
2		172.16.18.204	31.24K	48.20M	1.34%	100.00%	0.00%	00:00:01	1,125	0.00%
3		172.16.21.231	2.01K	34.10M	0.95%	3.87%	96.13%	00:09:35	575,937	0.02%
4		172.16.21.153	1.81K	30.61M	0.85%	3.20%	96.80%	00:08:33	513,197	0.02%
5		172.31.235.102	2.33K	28.89M	0.80%	5.85%	94.15%	00:21:18	1,278,828	0.05%
6		172.16.144.195	1.52K	21.46M	0.60%	6.45%	93.55%	00:10:07	607,122	0.02%
7		172.16.23.66	998	15.12M	0.42%	5.84%	94.16%	00:04:27	267,226	0.01%
8		172.16.23.207	667	11.71M	0.33%	3.21%	96.79%	00:03:18	198,418	0.01%
9		172.16.20.162	1.62K	10.94M	0.30%	0.32%	99.68%	00:13:24	804,194	0.03%
10		172.16.64.116	557	8.48M	0.24%	3.48%	96.52%	00:03:22	202,662	0.01%

Fig 3. Specific Date



NUM	ACCESSED SITE	CONNECT	BYTES	TIME	USERS
1	dm.mistat.com	31.24K	48.20M	0:00:01	1
2	172.16.16.36:9090	9.79K	157.22M	0:49:52	12
3	images.mid-day.com	4.50K	70.32M	0:18:21	2
4	pagead2.googlesyndication.com	3.86K	61.63M	0:14:08	26
5	maharashtratimes.indiatimes.com	3.03K	13.42M	0:08:59	1
6	www.google-analytics.com	2.63K	1.96M	0:07:41	82
7	googleads.g.doubleclick.net	2.63K	18.15M	0:17:34	76
8	www.mid-day.com	2.52K	47.09M	0:22:37	87
9	archive.mid-day.com	2.48K	53.65M	0:37:11	2
10	fbcdn-profile-a.akamaihd.net:443	2.36K	49.91M	91:51:04	1
11	b.scorecardresearch.com	2.29K	1.00M	0:03:41	82
12	safebrowsing-cache.google.com	2.04K	56.36M	0:10:21	1
13	www.juxtconsult.com	1.93K	1.23M	0:22:55	1
14	newmail.mid-day.com	1.84K	13.46M	0:30:09	12
15	economictimes.indiatimes.com	1.80K	3.77M	0:04:14	1
16	navbharattimes.indiatimes.com	1.75K	13.76M	0:08:27	1
17	epaper2.mid-day.com	1.65K	11.39M	0:35:33	9

Fig 4. Top Accessed Sites

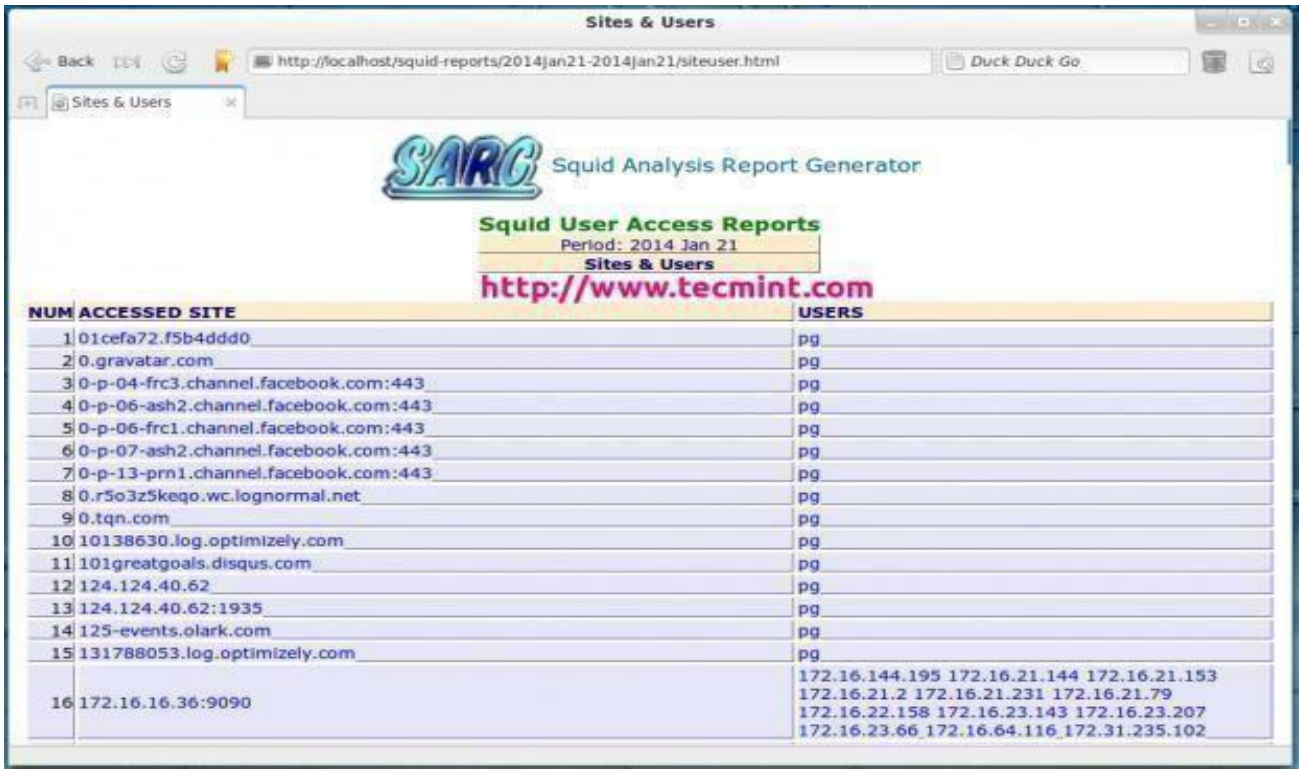


Fig 5. Top Sites and Users

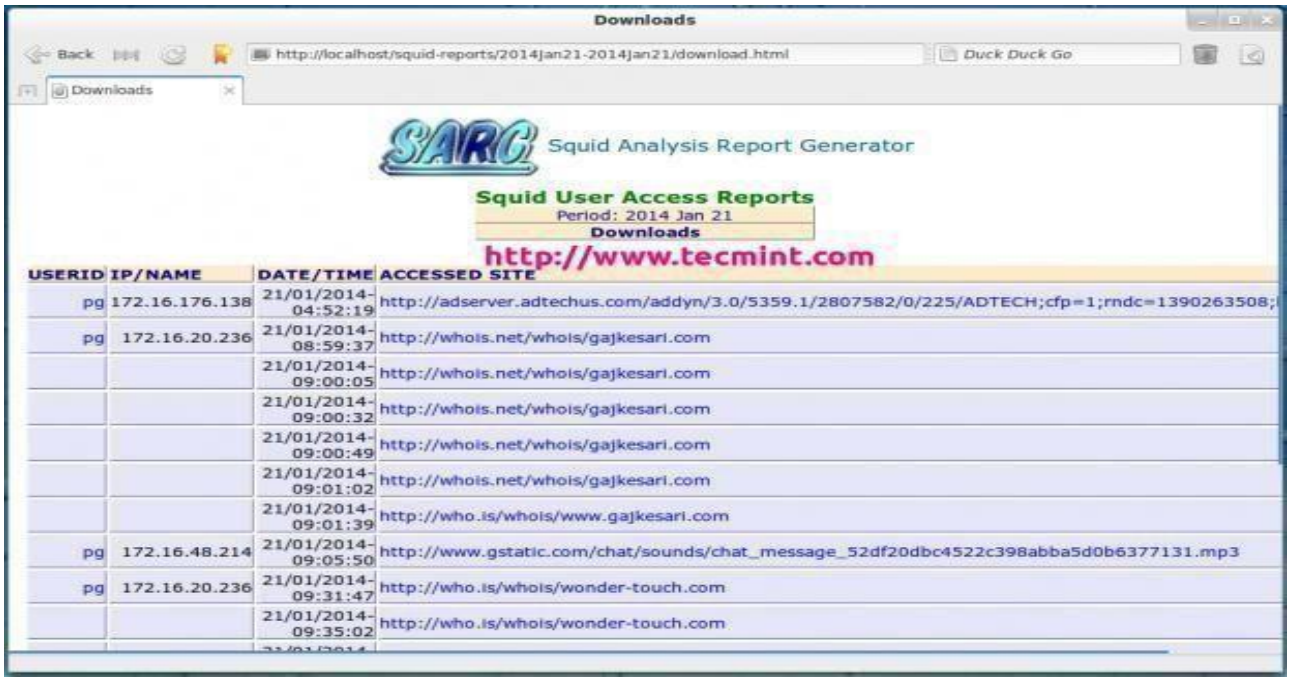


Fig 6. Top Downloads

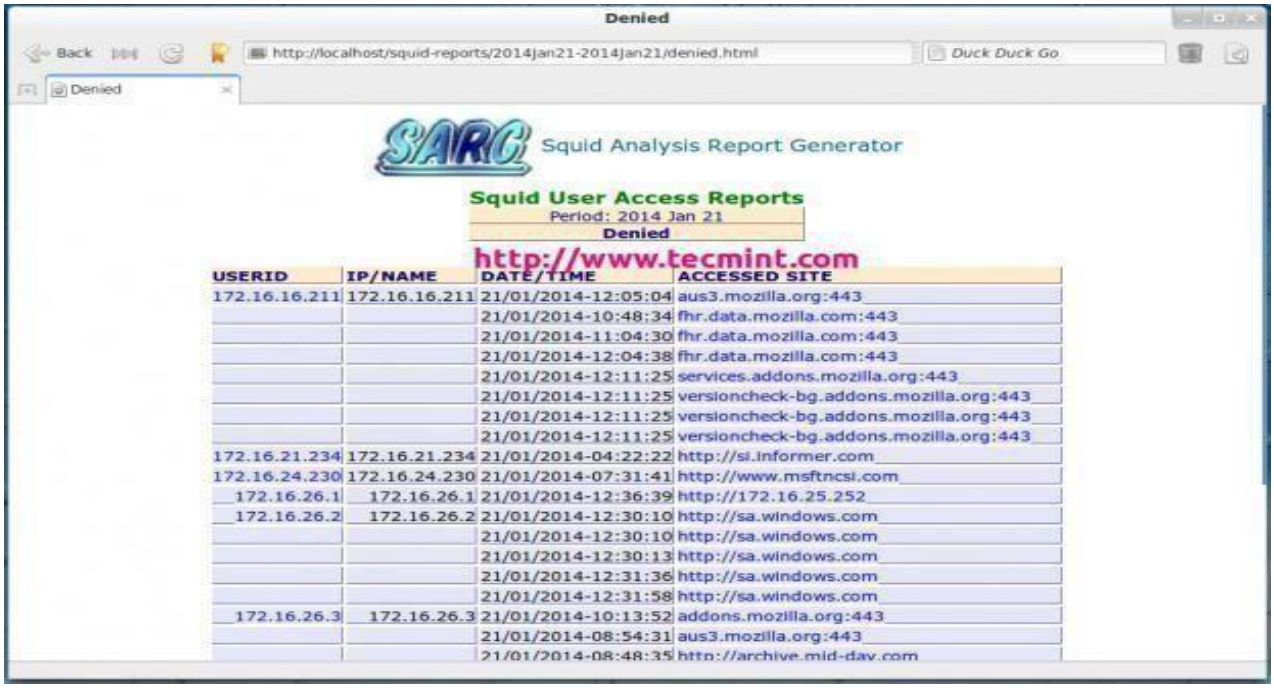


Fig 7. Denied Access

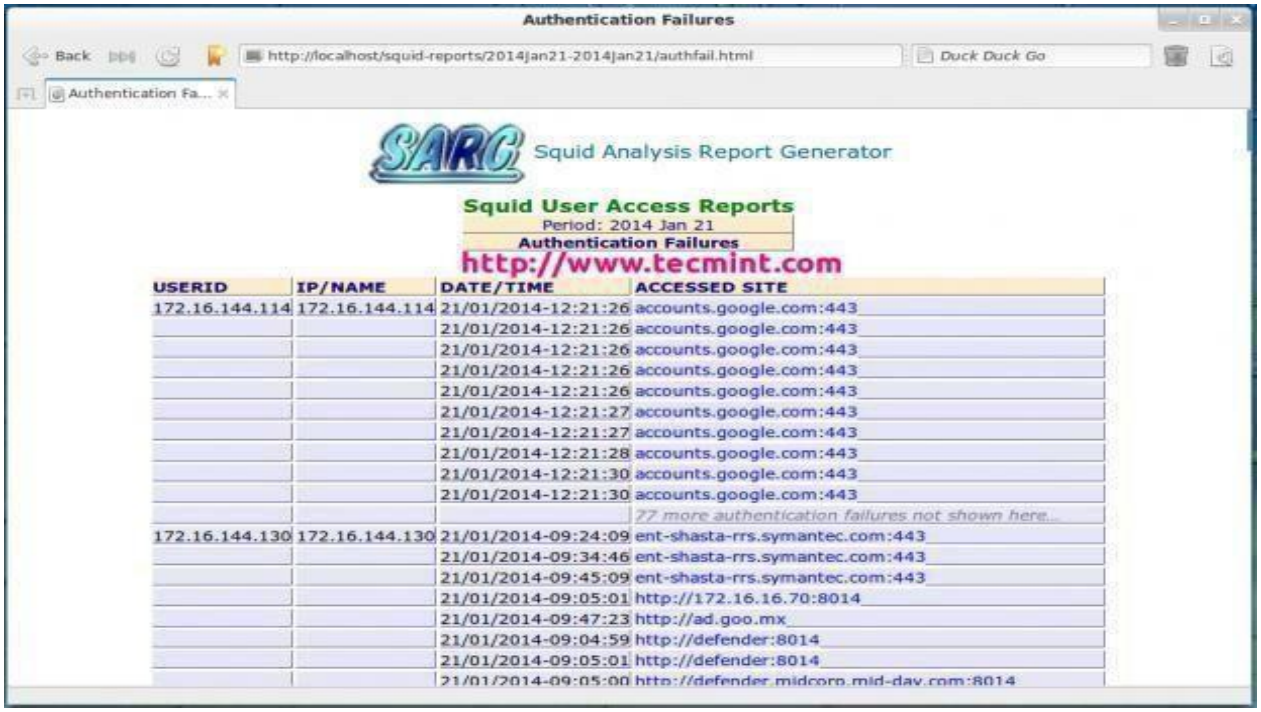


Fig 8. Authentication Failures

Assignment Group A-5

Problem Definition:

Design and Implement of Honeypot

Problem statement: Study and Implementation of Honeypot

Learning objective:

- To learn the concept of Honeypot
- To study the representation, implementation of Honeypot.

Learning outcome:

- Use honeybot tool to capture packets and configure tools and systems to enter unknown unauthenticated IP.

Software and hardware requirement:

- 64 bit machine
- Windows 7/8 Operating System

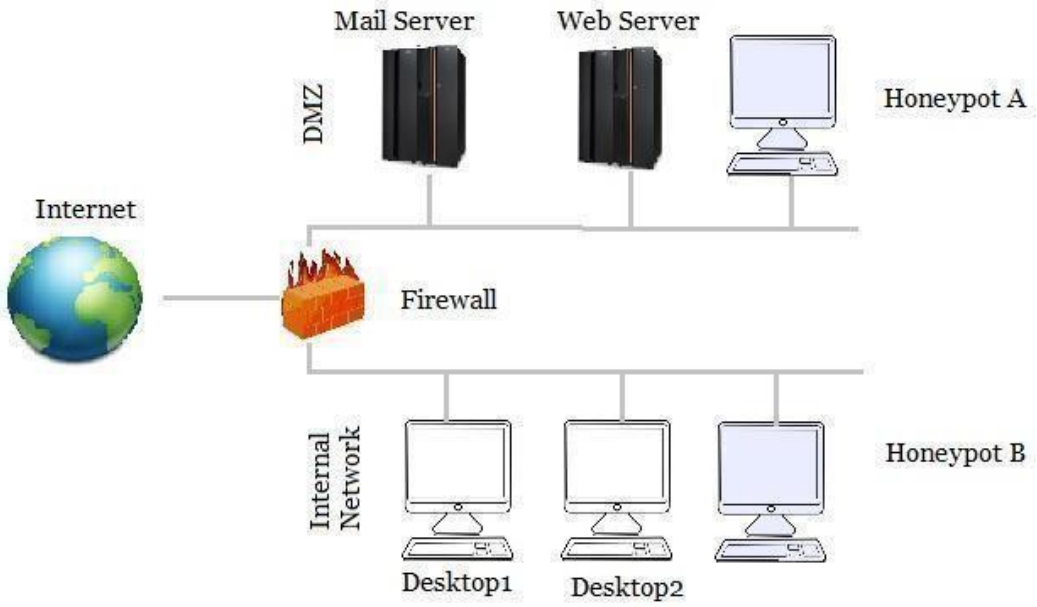
2. 1Theory:

Honeypot:

It is a computer system. There are files, directories in it just like a real computer. However, the aim of the computer is to attract hackers to fall into it to watch and follow their behavior. So we can define it as a fake system which looks like a real system. They are different than other security systems since they are not only finding one solution to a particular problem, but also they are eligible to apply variety of security problems and finding several approaches for them. For example, they can be used to log

Malicious activities in a compromised system; they can be also used to learn new threats for users and creating ideas how to get rid of those problems.

Honeypots are security resources that have no production value; no person or resource should be communicating with them. Any activity sent their way is suspect. Any traffic initiated by the honeypot means the system has most likely been compromised. Any traffic sent to the honeypot is most likely a probe, scan, or attack. With a honeypot, nothing is expected. To better understand the concepts of honeypots, let's take a look at the following example of honeypot deployments. Refer figure 11.1.



The purpose here is to demonstrate to you that honeypots can come in many different flavors, and they can achieve different things. However, they are both honeypots because they share the same definition and concepts. With the intent using systems as a honeypots, to determine if there is any unauthorized activity happening within your DMZ.

Honeypots passively capture any traffic or activity that interacts with them

5.1.1 Types of Honeypots:

There are two general types of honeypots:

5.1.1.1 **Production honeypots** are easy to use, capture only limited information, and are used primarily by companies or corporations. They are capturing a limited amount of information; mostly low interaction honeypots are used. security administrator watches the hacker's movements carefully and tries to lower the risks that may come from it towards the company

5.1.1.2 **Research honeypots** are complex to deploy and maintain, capture extensive information, and are used primarily by research, military, or government organizations. The objective is to learn how to Protect a system better, they do not bring any direct value to the security of an organization Honeypots are increasingly used to provide early warning of potential intruders, identify flaws in security strategies, and improve an organization's overall security awareness. "Honeypots can simulate a variety of internal and external devices, including Web servers, mail servers, database servers,

application servers, and even firewalls. As a software development manager, we can regularly use honeypots to gain insight into vulnerabilities in both the software my team writes and the OS upon which we depend."

A honeypot is a security resource whose value lies in being probed, attacked, or compromised. This means that whatever we designate as a honeypot, it is our expectation and goal to have the system probed, attacked, and potentially exploited.

5.2 Legal issues with honeypots

While deploying and start using a honeypot, there are some legal issues that a person should know about. Every country has different laws regarding to honeypot usage and information capturing. These regulations are related to data security, collection of data and finally how to use honeypots. All these different laws are based on the quality of the data that a honeypot can capture and a person who is deploying it. Privacy and data leads us to confidentiality term in network security. Our example is being a network administrator in a company.

5.3 Practical implementation

We are starting with low interaction honeypot and then continue on a middle level of interaction to finally conclude with a high level of interaction.

- **Starting to honeypot**,

We started with Honeyd as low level interaction honeypot and then we will move on medium level interaction honeypots. Every honeypot has specific and different attitudes. We will explain them one by one.

- **HoneyBOT is a medium interaction honeypot for windows.**

A honeypot creates a safe environment to capture and interact with unsolicited and often malicious traffic on a network. HoneyBOT is an easy to use solution ideal for network security research or as part of an early warning IDS. The logging capability of a honeypot is far greater than any other network security tool and captures raw packet level data even including the keystrokes and mistakes made by hackers. The captured information is highly valuable as it contains only malicious traffic with little to no false positives. Honeypots are becoming one of the leading security tools used to monitor the latest tricks and exploits of hackers by recording their every move so that the security community can more quickly respond to new exploits.

- **How does it work?**

HoneyBOT works by opening a range of listening sockets on your computer which are designed to mimic vulnerable services. When an attacker connects to these services they are fooled into thinking they are attacking a real server. The honeypot safely captures all communications with the attacker and logs these results for future analysis. Should an attacker attempt an exploit or upload a rootkit or trojan to the server the honeypot environment can safely store these files on your computer for malware collection and analysis purposes. Following figure shows implementation of honeypot.

- **Installing and Securing Your Honeypot**

A honeypot is intentionally put in harms way so it is critical to carry out some security precautions on your honeypot computer before deployment on any network. Install HoneyBOT on a dedicated computer or virtual machine. Update the operating system with security updates and use an antivirus product. You want your honeypot to be as free as possible from legitimate traffic so in broad terms we can consider any traffic to the honeypot to be malicious in nature. Remember that we are attracting attackers to intrude into this system so precautions are important.

- **Network Placement**

If you place HoneyBOT inside the internal network where it is secured by perimeter defences it should never to be attacked. Any malicious traffic captured in this situation would indicate that another computer inside the network is already compromised or that the perimeter defences have been breached. In this configuration HoneyBOT is acting as an intrusion detection system. If you place HoneyBOT on an external network or internet you will attract higher volumes of unsolicited network traffic. Direct internet placement is the most common setup with HoneyBOT being on the network DMZ.

- **Windows Services, SMB and NetBIOS**

You should disable any Windows services that are not required for the machine to operate as they offer an attacker a possible avenue of attack. HoneyBOT cannot listen on a port that is already in use by a Windows service. Some of the services that you may choose to disable include Messenger, ClipBook, COM+, FTP Publishing, SMTP, SNMP, TCP/IP NetBIOS Helper, Telnet, WWW Publishing.

SMB (CIFS) provides name resolution, network browsing and printing services over TCP/IP. To disable SMB open the Network Connections window, right click the adapter and select Properties and uninstall Client For Microsoft Networks and File And Printer Sharing. SMB services may also be provided over NetBIOS (NBT). To disable NetBIOS open the Device Manager window, select Show Hidden Devices, expand Non-Plug And Play Drivers and disable NetBios Over Tcpip.

If you are monitoring your honeypot via a remote desktop tool then you should change the default listening port to a random high numbered port.

Finally, before starting HoneyBOT take a baseline of the current listening services by opening a command shell and launching netstat with the -ano option. Any listening services that you are unable to disable need to be blocked at the firewall.

- **Firewall**

A firewall will prevent unsolicited connections from reaching your computer. In order for HoneyBOT to communicate you need to customise your firewall rules to allow incoming connections. If you are using a software firewall you should create an exception for HoneyBOT.

- **HoneyBOT Options**

Select Options from the View menu to configure HoneyBOT.

Automatically Start Engine: The server engine will start automatically when the application is started.

Enable Sound Alert: Plays a short sound each time an event occurs. Capture Binaries: If this option is enabled HoneyBOT will attempt to capture malware and other files and save them to the \HoneyBOT\Captures\ folder. If this option is enabled you should add an exception in your antivirus software to exclude this folder from its scan. Automatically Rotate

Log: Each day at midnight HoneyBOT will save the current log file and start a new log file. Server Name: The alias name of the HoneyBOT server given to the remote machine.

- **Email Alerts**

Enter your email address and SMTP server information to receive daily email updates from HoneyBOT.

- **Exports**

Select the Export Logs to CSV option to create a daily extract of your log file as a CSV file. Exported logs are saved in the \HoneyBOT\Logs\ folder. You can also choose to participate in the centralised log program and have your log files uploaded to the HoneyBOT website.

- **Syslog**

Select to send connection events to a Syslog server. Enter the Syslog server IP address and

port.

- **Bindings**

Only applicable to multihomed machines. Provides support for multiple networks so HoneyBOT can bind to one or all detected networks. Enter the IP address that you want HoneyBOT to bind to. If the IP address is not valid and more than one IP address is available you will be prompted to select an address when the server engine starts.

- **Updates**

Select to have HoneyBOT check for updates on startup. There are two update types that may occur. A service update is a minor update to the server listening services, if a service update is available you will be prompted to install the update. An application update notification will occur if a new version of HoneyBOT is available.

- **Services and Profiles**

Select to edit the TCP and UDP services started by the HoneyBOT engine. You can add a new port, edit and disable an existing port, or delete the port configuration entirely.

By default HoneyBOT will open more listening ports than a typical computer and this may alert an attacker to its presence. You can choose to limit your honeypot exposure to just a handful of ports that more closely resembles a real operating system. By loading a profile you can quickly emulate common operation system setups like an SQL Server, IIS Server, Exchange Server, etc.

- **Whitelist**

You may find HoneyBOT is interacting with services on your network that are legitimate and not a cause for alarm. You can whitelist the source machine by adding the IP and port to the whitelist settings. When a machine is whitelisted HoneyBOT will no longer accept connections from that machine.

- **Debug**

The debug window will display application messages and socket events that occur during typical application operation.

- **Event Navigation**

The event tree on the left shows the ports that have been probed and remote addresses

that have connected to HoneyBOT. The event list at the top right will display all connection attempts including the attributes of the connection. The packet list at the bottom displays each packet transmitted and received between the remote machine and the HoneyBOT server. You can expand the event tree and filter the events displayed by selecting an item in the list.

5.4 Advantages of honeypots

There are many security solutions available in the market. Anyone can browse the variety of choices through internet and find the most suitable solution for their needs. Honeypots can capture attacks and give information about the attack type and if needed, thanks to the logs, it is possible to see additional information about the attack. New attacks can be seen and new security solutions can be created by looking at them. More examinations can be obtained by looking at the type of the malicious behaviors. It helps to understand more attacks that may happen. Honeypots are not bulky in terms of capturing data. They are only dealing with the incoming malicious traffic. Therefore, the information that has been caught is not as much as the whole traffic. Focusing only on the malicious traffic makes the investigation far easier.

5.5 Disadvantages of honey pots

We can only capture data when the hacker is attacking the system actively. If he does not attack the system, it is not possible to catch information. If there is an attack occuring in another system, our honeypot will not be able to identify it. So, attacks not towards our honeypot system may damage other systems and cause big problems. There is fingerprinting disadvantage of honeypots. It is easy for an experienced hacker to understand if he is attacking a honeypot system or a real system. Fingerprinting allows us to distinguish between these two. It is a not a wanted result of our experiment. The honeypot may be used as a zombie to reach other systems and compromise them. This can be very dangerous.

Experiment No: Group B-1

Problem Definition:

To implement a basic function of Code Division Multiple Access (CDMA) to test the orthogonally and autocorrelation of a code to be used for CDMA operation. Write an application based on the above concept.

1.1 Prerequisite:

Basic Concept of Code Division Multiple Access

1.2 Learning Objective:

1. To provides a high quality of voice with almost no noise during the calls.
2. To Understand working of CDMA through common channel.

1.3 Theory:

1.3.1 Introduction

CDMA is a channelization protocol for Multiple Access, where information can be sent simultaneously through several transmitters over a single communication channel.

It is achieved in below steps:

- A signal is generated which extends over a wide bandwidth.
- The code which performs this action is called spreading code.
- Later on, a specific signal can be selected with a given code even in the presence of

many other signals.

- It is mainly used in mobile networks like 2G and 3G.

1.3.2 How does CDMA work?

To see how CDMA works, we have to understand orthogonal sequences (also known as chips).

Let N be the number of stations establishing multiple access over a common channel.

Then the properties of orthogonal sequences can be stated as follows:

1. An orthogonal sequence can be thought of as a 1xN matrix.

   Eg: [+1 -1 +1 -1] for N = 4.

2. Scalar multiplication and matrix addition rules follow as usual.

   Eg: 3.[+1 -1 +1 -1] = [+3 -3 +3 -3]

   Eg: [+1 -1 +1 -1] + [-1 -1 -1 -1] = [0 -2 0 -2]

3. Inner Product: It is evaluated by multiplying two sequences element by element and then adding all elements of the resulting list.
   - Inner Product of a sequence with itself is equal to N

     [+1 -1 +1 -1].[+1 -1 +1 -1] = 1 + 1 + 1 + 1 = 4
   - Inner Product of two distinct sequences is zero

     [+1 -1 +1 -1].[+1 +1 +1 +1] = 1-1+1-1 = 0
   - To generate valid orthogonal sequences, use a Walsh Table as follows:
   - Rule 1:

     $$W_1 = [+1]$$
   - Rule 2:

     $$W_{2N} = \begin{bmatrix} W_N & W_N \\ W_N & \overline{W_N} \end{bmatrix}$$

Where $\overline{W_N}$ = Complement of W_N (Replace +1 by -1 and -1 by +1)

Example:

$$W_2 = \begin{bmatrix} +1 & +1 \\ +1 & -1 \end{bmatrix}$$

$$W_4 = \begin{bmatrix} +1 & +1 & +1 & +1 \\ +1 & -1 & +1 & -1 \\ +1 & +1 & -1 & -1 \\ +1 & -1 & -1 & +1 \end{bmatrix}$$

- Each row of the matrix represents an orthogonal sequence.Hence we can construct sequences for N = 2^M. Now let's take a look at how CDMA works by using orthogonal sequences.

1.3.3 Procedure:

1. The station encodes its data bit as follows.
    - +1 if bit = 1
    - -1 if bit = 0
    - no signal(interpreted as 0) if station is idle
2. Each station is assigned a unique orthogonal sequence (code) which is N bit long for N stations
3. Each station does a scalar multiplication of its encoded data bit and code sequence.
4. The resulting sequence is then placed on the channel.
5. Since the channel is common, amplitudes add up and hence resultant channel sequence is sum of sequences from all channels.
6. If station 1 wants to listen to station 2, it multiplies (inner product) the channel

sequence with code of station S2.

7. The inner product is then divided by N to get data bit transmitted from station 2.

1.3.4 :Example: Assume 4 stations S1, S2, S3, S4. We'll use 4×4 Walsh Table to assign codes to them.

C1 = [+1 +1 +1 +1]

C2 = [+1 -1 +1 -1]

C3 = [+1 +1 -1 -1]

C4 = [+1 -1 -1 +1]

Let their data bits currently be:

D1 = -1

D2 = -1

D3 = 0 (Silent)

D4 = +1

Resultant channel sequence = C1.D1 + C2.D2 + C3.D3 + C4.D4

= [-1 -1 -1 -1] + [-1 +1 -1 +1] + [0 0 0 0]+[+1 -1 -1 +1]

= [-1 -1 -3 +1]

Now suppose station 1 wants to listen to station 2.

Inner Product = [-1 -1 -3 +1] x C2

= -1 + 1 - 3 - 1 = -4

Data bit that was sent = -4/4 = -1.

1.3.5 Code:

```
import numpy as np

c1=[1,1,1,1]

c2=[1,-1,1,-1]

c3=[1,1,-1,-1]

c4=[1,-1,-1,1]

rc=[]


print("Enter the data bits :")

d1=int(input("Enter D1 :"))

d2=int(input("Enter D2 :"))

d3=int(input("Enter D3 :"))

d4=int(input("Enter D4 :"))

r1=np.multiply(c1,d1)

r2=np.multiply(c2,d2)

r3=np.multiply(c3,d3)

r4=np.multiply(c4,d4)

resultant_channel=r1+r2+r3+r4;
```

```
print("Resultant Channel",resultant_channel)

Channel=int(input("Enter  the station to listen for C1=1 ,C2=2, C3=3 C4=4 : "))

if Channel==1:

        rc=c1

elif Channel==2:

        rc=c2

elif Channel==3:

        rc=c3

elif Channel==4:

        rc=c4

inner_product=np.multiply(resultant_channel,rc)

print("Inner Product",inner_product)

res1=sum(inner_product)

data=res1/len(inner_product)

print("Data bit that was sent",data)
```

Assignment No.:- 1

Write TEST Scenario for Gmail Login Page

Test Cases – Login Page

Following is the possible list of functional and non-functional test cases for a login page:

Functional Test Cases:

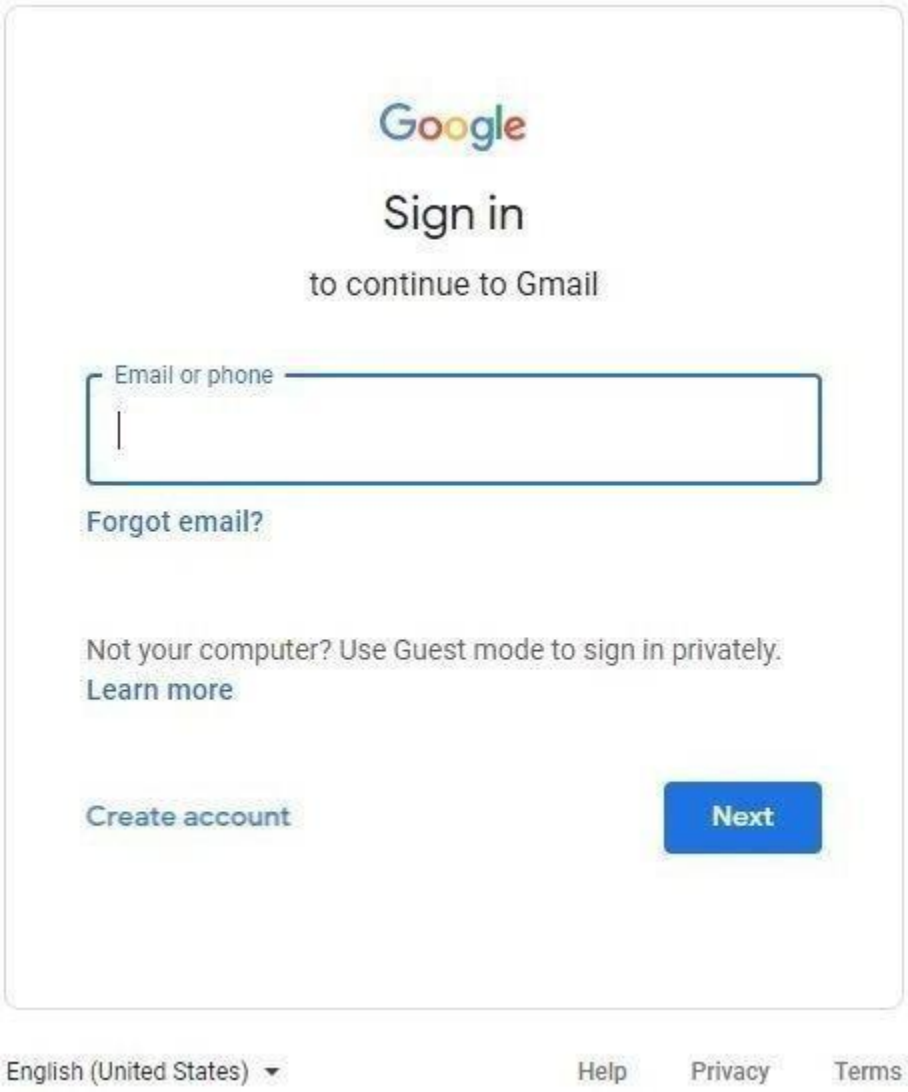
Sr. No.	Functional Test Cases	Type- Negative/ Positive Test Case
1	Verify if a user will be able to login with a valid username and valid password.	Positive
2	Verify if a user cannot login with a valid username and an invalid password.	Negative
3	Verify the login page for both, when the field is blank and Submit button is clicked.	Negative
4	Verify the 'Forgot Password' functionality.	Positive
5	Verify the messages for invalid login.	Positive
6	Verify the 'Remember Me' functionality.	Positive
7	Verify if the data in password field is either visible as asterisk or bullet signs.	Positive
8	Verify if a user is able to login with a new password only after he/she has changed the password.	Positive
9	Verify if the login page allows to log in simultaneously with different credentials in a different browser.	Positive
10	Verify if the 'Enter' key of the keyboard is working correctly on the login page.	Positive
	Other Test Cases	
11	Verify the time taken to log in with a valid username and password.	Performance & Positive Testing
12	Verify if the font, text color, and color coding of the Login page is as per the standard.	UI Testing & Positive Testing
13	Verify if there is a 'Cancel' button available to erase the entered text.	Usability Testing
14	Verify the login page and all its controls in different browsers	Browser Compatibility & Positive Testing.

Non-functional Security Test Cases:

Sr. No.	Security test cases	Type- Negative/ Positive Test Case
1	Verify if a user cannot enter the characters more than the specified range in each field (Username and Password).	Negative
2	Verify if a user cannot enter the characters more than the specified range in each field (Username and Password).	Positive
3	Verify the login page by pressing ‘Back button’ of the browser. It should not allow you to enter into the system once you log out.	Negative
4	Verify the timeout functionality of the login session.	Positive
5	Verify if a user should not be allowed to log in with different credentials from the same browser at the same time.	Negative
6	Verify if a user should be able to login with the same credentials in different browsers at the same time.	Positive
7	Verify the Login page against SQL injection attack.	Negative
8	Verify the implementation of SSL certificate.	Positive

We can take an Example of Gmail Login page. Here is the image of it.

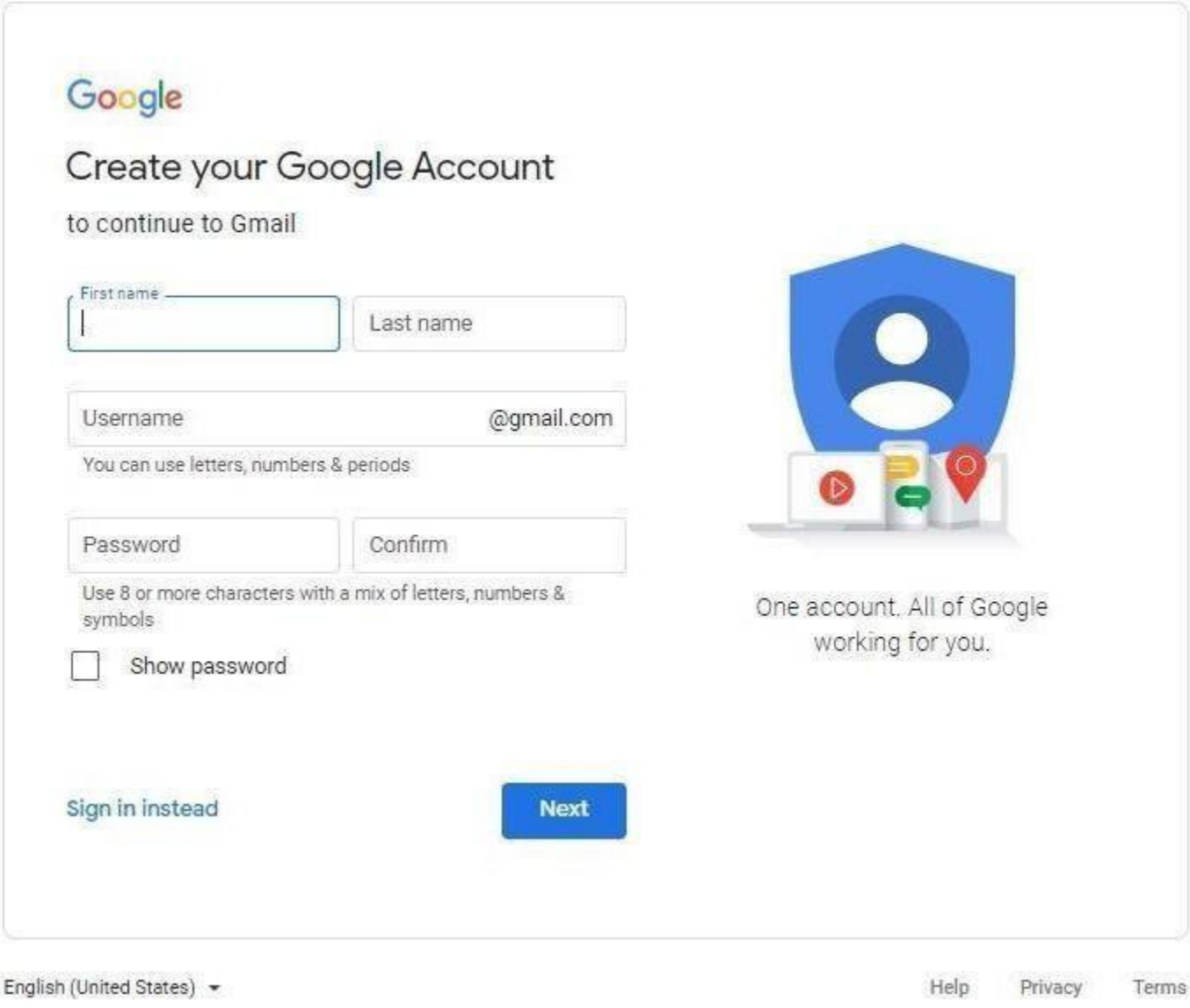
Test Cases for Gmail Login page



Sr. No.	Test Scenarios
1	Enter the valid email address & click next. Verify if the user gets an option to enter the password.
2	Don’t enter an email address or phone number & just click the Next button. Verify if the user will get the correct message or if the blank field will get highlighted.
3	Enter the invalid email address & click the Next button. Verify if the user will get the correct message.
4	Enter an invalid phone number & click the Next button. Verify if the user will get the correct message.
5	Verify if a user can log in with a valid email address and password.
6	Verify if a user can log in with a valid phone number and password.

Sr. No.	Test Scenarios
7	Verify if a user cannot log in with a valid phone number and an invalid password.
8	Verify if a user cannot log in with a valid email address and a wrong password.
9	Verify the ‘Forgot email’ functionality.
10	Verify the ‘Forgot password’ functionality.

Test Scenarios for the Sign-up page



Test Scenarios for the Sign-up page

1) Verify the messages for each mandatory field.

2) Verify if the user cannot proceed without filling all the mandatory fields.

3) Verify the age of the user when the DOB is selected.

4) Verify if the numbers and special characters are not allowed in the First and Last name.

5) Verify if a user can sign-up successfully with all the mandatory details.

6) Verify if a user can log in with the valid details.

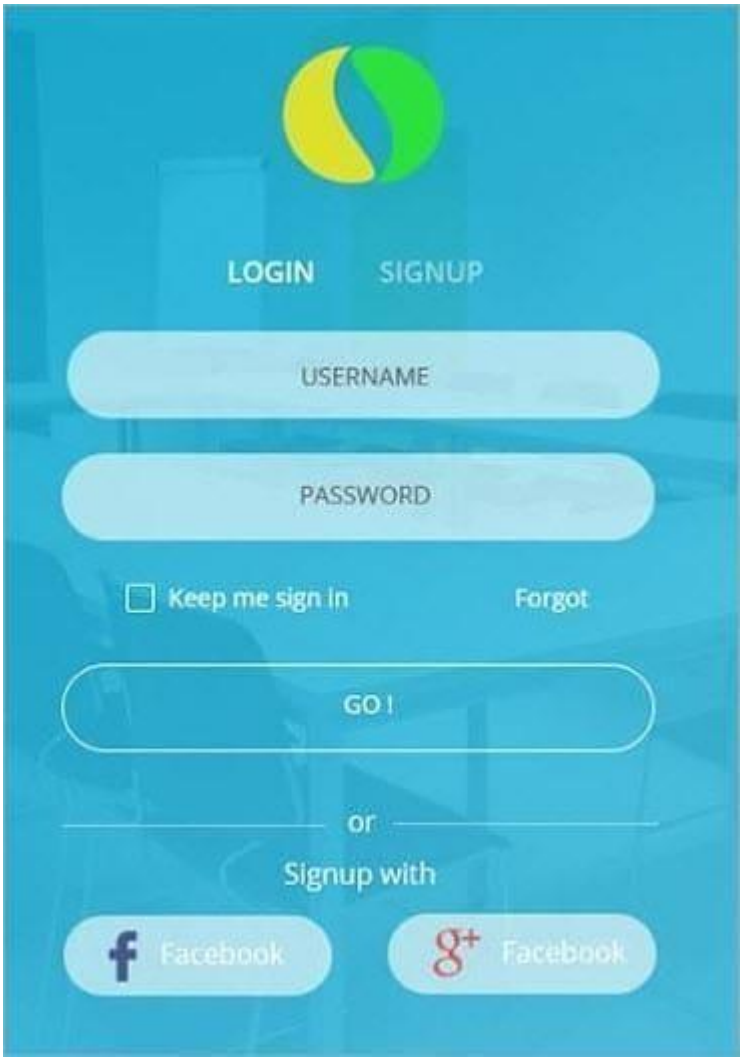
7) Verify if the Password and Confirm Password fields are accepting similar strings only.

8) Verify if the Password field will prompt you for the weak passwords.

9) Verify if duplicate email address will not get assigned.

10) Verify that hints are provided for each field on the form, for the ease of use.

Test Scenarios for the Login page of Mobile Application



1) Verify if a user can log in with a valid username and password.

2) Verify if a user cannot log in with an invalid username or password. Check permutation and combinations of this.

3) Verify the 'Keep me Sign In' option. If this check box is selected, then the user should not get logged out even after exiting the app.

4) Verify if this check box is not selected by default.

5) If the user has signed up with Facebook or social media, verify that the user can log in with those credentials or not.

6) Verify the Forgot password functionality.

7) Verify if the login page fits the mobile screen. The user should not have to scroll the screen.

Conclusion

While writing test cases for login or sign-up page write the test cases for all the fields. There should be a combination of both positive and negative test cases. Try to cover the performance, security, and functional scenarios.

The login page is the page with fewer controls, so even though it is looking simple for testing, it should not be considered as an easy task.

Also many a time it is the first impression of an application, so it should be perfect for user interface and usability.

Assignment No.:- 2

Test Scenario for Gmail – Inbox Functionality

Test Case for Gmail – Inbox Functionality

1) Verify that a newly received email is displayed as highlighted in the Inbox section.
2) Verify that a newly received email has correctly displayed sender email Id or name, mail subject and mail body(trimmed to a single line).
3) Verify that on clicking the newly received email, the user is navigated to email content.
4) Verify that the email contents are correctly displayed with the desired source formatting.
5) Verify that any attachments are attached to the email and are downloadable.
6) Verify that the attachments are scanned for viruses before download.
7) Verify that all the emails marked as read are not highlighted.
8) Verify that all the emails read as well as unread have a mail read time appended at the end on the email list displayed in the inbox section.
9) Verify that count of unread emails is displayed alongside 'Inbox' text in the left sidebar of Gmail.
10) Verify that unread email count increases by one on receiving a new email.
11) Verify that unread email count decreases by one on reading an email (marking an email as read).
12) Verify that email recipients in cc are visible to all users.
13) Verify that email recipients in bcc are not visible to the user.
14) Verify that all received emails get piled up in the 'Inbox' section and get deleted in cyclic fashion based on the size availability.
15) Verify that email can be received from non-Gmail email Ids like – yahoo, Hotmail etc.

Test Cases for GMail – Compose Mail Functionality

1) Verify that on clicking 'Compose' button, a frame to compose a mail gets displayed.
2) Verify that user can enter email Ids in 'To', 'cc' and 'bcc' sections and also user will get suggestions while typing the emailds based on the existing emailIds in user's email list.
3) Verify that the user can enter multiple comma-separated emailIds in 'To', 'cc' and 'bcc' sections.
4) Verify that the user can type Subject line in the 'Subject' textbox.
5) Verify that the user can type the email in the email-body section.
6) Verify that users can format mail using editor-options provided like choosing font-family, font-size, bold-italic-underline, etc.
7) Verify that the user can attach file as an attachement to the email.
8) Verify that the user can add images in the email and select the size for the same.
9) Verify that after entering emailIds in either of the 'To', 'cc' and 'bcc' sections, entering Subject line and mail body and clicking 'Send' button, mail gets delivered to intended receivers.
10) Verify that sent mails can be found in 'Sent Mail' sections of the sender.

11) Verify that mail can be sent to non-gmail emailIds also.
12) Verify that all sent emails get piled up in the 'Sent Mail' section and get deleted in cyclic fashion based on the size availability.
13) Verify that the emails composed but not sent remain in the draft section.
14) Verify the maximum number of email recipients that can be entered in 'To', 'cc' and 'bcc' sections.
15) Verify the maximum length of text that can be entered in the 'Subject' textbox.
16) Verify the content limit of text/images that can be entered and successfully delivered as mail body.
17) Verify the maximum size and number of attachment that can be attached with an email.
18) Verify that only the allowed specifications of the attachment can be attached with an email/
19) Verify that if the email is sent without Subject, a pop-up is generated warning user about no subject line. Also,
20) Verify that on accepting the pop-up message, the user is able to send the email.

Write Test cases in excel sheet for Social Media application or website

Sample Test Case Template With Test Case Examples

Every day I keep on getting several requests for a Test Case Template. I'm surprised that many testers are still documenting test cases with Word docs or Excel files.

Most of them prefer excel spreadsheets because they can easily group test cases by test types and most importantly they can easily get test metrics with Excel formulas. But I'm sure that as the volume of your tests goes on increasing, you will find it extremely difficult to manage.

If you are not using any Test case management tool, then I would strongly recommend you to use an open-source tool to manage and execute your test cases.

Test case formats may vary from one organization to another. However, using a standard test case format for writing test cases is one step closer to setting up a testing process for your project.

It also minimizes Ad-hoc testing that is done without proper test case documentation. But even if you use standard templates, you need to set up test cases writing, review & approve, test execution and most importantly test report preparation process, etc. by using manual methods.

Also, if you have a process to review the test cases by the business team, then you must format these test cases in a template that is agreed by both the parties.

Recommended Tools

Before continuing with the Test case writing process, we recommend downloading these Test case management tools. This will ease your test plan and test case writing process mentioned in this tutorial.

1) TestRail

TestRail is a web-based tool for test cases and test management. It helps QA and development teams with the efficient management of test cases, plans, and runs. It gives centralized test management, powerful reports & metrics, and increased productivity. It is a scalable and customizable solution. It can be used by small as well as large teams.

Features:

1) TestRail makes tracking test results easier.
2) It seamlessly gets integrated with bug trackers, automated tests, etc.
3) Personalized to-do lists, filters, and email notifications will help with boosting productivity.
4) Dashboards and activity reports are for easy tracking and following the status of individual tests, milestones, and projects.

2) Katalon Studio

Katalon Studio is an all-in-one, simple automation tool for web, API, mobile, and desktop trusted by over 850,000 users.

It simplifies automation for those without a coding background to create automation test cases from manual tests' steps, a rich library of project templates, record & playback, and a friendly UI.

3) Testiny

Testiny – a new, straightforward test management tool, but much more than just a slimmed-down app.

Testiny is a fast-growing web application built on the latest technologies and aims to make manual testing and QA management as seamless as possible. It is designed to be extremely easy to use. It helps testers perform tests without adding bulky overhead to the testing process.

Don't just take our word for it, take a look at Testiny yourself. Testiny is perfect for small to mid-sized QA teams looking to integrate manual and automated testing into their development process.

Features:

1) Free for open-source projects and small teams with up to 3 people.
2) Intuitive and simple out of the box.
3) Easily create and handle your test cases, test runs, etc.
4) Powerful integrations (e.g. Jira, …)
5) Seamless integration in the development process (linking requirements and defects)
6) Instant updates – all browser sessions stay in sync.
7) Immediately see if a colleague has made changes, completed a test, etc.
8) Powerful REST API.
9) Organize your tests in a tree structure – intuitive and easy.

Standard Fields of a Sample Test Case Template

There are certain standard fields that need to be considered while preparing a Test case template.

Project Name:	
Test Case Template	
Test Case ID: Fun_10	**Test Designed by:** <Name>
Test Priority (Low/Medium/High): Med	**Test Designed date:** <Date>
Module Name: Google login screen	**Test Executed by:** <Name>
Test Title: Verify login with valid username and	**Test Execution date:** <Date>

password	
Description: Test the Google login page	
Pre-conditions: User has valid username and password	
Dependencies:	

Step	Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)	Notes
1	Navigate to login page	User= example@gmail.com	User should be able to login	User is navigated to	Pass	
2	Provide valid username	Password: 1234		dashboard with successfu		
3	Provide valid password			Login		
4	Click on Login button					

Post-conditions:
User is validated with database and successfully login to account. The account session details are logged in database.

Several standard fields for a sample Test Case template are listed below.

1) Test case ID: Unique ID is required for each test case. Follow some conventions to indicate the types of the test. For Example, 'TC_UI_1' indicating 'user interface test case #1'.

2) Test priority (Low/Medium/High): This is very useful during test execution. Test priorities for business rules and functional test cases can be medium or higher, whereas minor user interface cases can be of a low priority. Testing priorities should always be set by the reviewer.
3) Module Name: Mention the name of the main module or the sub-module.
4) Test Designed By: Name of the Tester.
5) Test Designed Date: Date when it was written.
6) Test Executed By Name of the Tester who executed this test. To be filled only after test execution.
7) Test Execution Date: Date when the test was executed.
8) Test Title/Name: Test case title. For example, verify the login page with a valid username and password.
9) Test Summary/Description: Describe the test objective in brief.
10) Pre-conditions: Any prerequisite that must be fulfilled before the execution of this test case. List all the pre-conditions in order to execute this test case successfully.
11) Dependencies: Mention any dependencies on other test cases or test requirements.

12) Test Steps: List all the test execution steps in detail. Write test steps in the order in which they should be executed. Make sure to provide as many details as you can.
13) Test Data: Use of test data as an input for this test case. You can provide different data sets with exact values to be used as an input.
14) Expected Result: What should be the system output after test execution? Describe the expected result in detail including the message/error that should be displayed on the screen.
15) Post-condition: What should be the state of the system after executing this test case?
16) Actual result: The actual test result should be filled after test execution. Describe the system behavior after test execution.
17) Status (Pass/Fail): If the actual result is not as per the expected result, then mark this test as failed. Otherwise, update it as passed.
18) Notes/Comments/Questions: If there are any special conditions to support the above fields, which can't be described above or if there are any questions related to expected or actual results then mention them here.

Add the following fields if necessary:

1) Defect ID/Link: If the test status fails, then include the link to the defect log or mention the defect number.

2) Test Type/Keywords: This field can be used to classify tests based on test types. For Example, functional, usability, business rules, etc.

3) Requirements: Requirements for which this test case is being written for. Preferably the exact section number in the requirement doc.
4) Attachments/References: This field is useful for complex test scenarios in order to explain the test steps or expected results using a Visio diagram as a reference. Provide a link or location to the actual path of the diagram or document.
5) Automation? (Yes/No): Whether this test case is automated or not. It is useful to track automation status when test cases are automated.

One More Test Case Format (#2)

The test cases will differ depending upon the functionality of the software that it is intended for. However, given below is a template that you can always use to document the test cases without bothering about what your application is doing?

Test Scenario ID		Test Case ID		
Test Case Description		Test Priority		
Pre-Requisite		Post-Requisite		
Test Execution Steps:				

S.No	Action	Inputs	Expected Output	Actual Output	Test Browser	Test Result	Test Comments

Sample Test Cases

Based on the above template, below is an example that showcases the concept in a much understandable way.

Let's assume that you are testing the login functionality of any web application, say Facebook.

An ideal test case template

Below is a template which you can always use for documenting the test cases without bothering with what your application is doing.

Test Scenario ID		**Test Case ID**		
Test Case Description		**Test Priority**		
Pre-Requisite		**Post-Requisite**		
Test Execution Steps:				

S.No	**Action**	**Inputs**	**Expected Output**	**Actual Output**	**Test Browser**	**Test Result**	**Test Comments**

Example Scenario

Based on the above template, below is an example that showcases the concepts in a more understandable way.

Suppose you are testing the login functionality of any web application, say Facebook. Below are the test cases for the same:

Test Scenario ID	Login-1	**Test Case ID**	Login-1A
Test Case Description	Login – Positive test case	**Test Priority**	High
Pre-Requisite	A valid user account	**Post-Requisite**	NA

Test Execution Steps:

S.No	**Action**	**Inputs**	**Expected Output**	**Actual Output**	**Test Browser**	**Test Result**	**Test Comments**
1	Launch application	https://www.facebook.com/	Facebook home	Facebook home	IE -11	Pass	[Priya 12/17/2016 11:44 AM]: Launch successful
2	Enter correct Email & Password and hit login button	Email id : test@xyz.com Password: ******	Login success	Login success	IE -11	Pass	[Priya 12/17/2016 11:45 AM]: Login successful

Test Scenario ID	Login-1	**Test Case ID**	Login-1B
Test Case Description	Login – Negative test case	**Test Priority**	High
Pre-Requisite	NA	**Post-Requisite**	NA

Test Execution Steps:

S.No	**Action**	**Inputs**	**Expected Output**	**Actual Output**	**Test Browser**	**Test Result**	**Test Comments**
1	Launch	https://www.facebo	Facebook	Faceboo	IE -11	Pass	[Priya 12/17/2016

	applicatio n	ok.com/	home	k home			11:44 AM]: Launch successful
2	Enter invalid Email & any Password and hit login button	Email id : invalid@xyz.com Password: ******	The email address or phone number that you've entered doesn't match any account. Sign up for an account.	The email address or phone number that you've entered doesn't match any account. Sign up for an account.	IE -11	Pass	[Priya 12/17/2016 11:45 AM]: Invalid login attempt stopped
3	Enter valid Email & incorrect Password and hit login button	Email id : valid@xyz.com Password: ******	The password that you've entered is incorrect. Forgotten password ?	The passwor d that you've entered is incorrect . Forgotte n passwor d?	IE -11	Pass	[Priya 12/17/2016 11:46 AM]: Invalid login attempt stopped

Samples from a Live Project

Below is examples from a live project that demonstrates how all the above listed tips and tricks are actually implemented:

	Test Scenario Group	Test Case Id	Test Case Description	Test Env	Test Input	Expected Result	Actual Result	Test Browser	Executed Date	Test Results	Executed Date	Test Results	Defect Status
2	Search	Search-1	Search for Patient only in Copia										
3	Search	Search-2	Search for Patient only in IMS										
4	Search	Search-3	Search for Patient in both Copia & IMS										
5	Search	Search-4	Search by first name										
6	Search	Search-5	Search by last name										
7	Search	Search-6	Search by patient id										
8	Search	Search-7	Search with partial name										
9	Search	Search-8	Search with partial patient id										
10	Search	Search-9	Update Patient profile										
11			a) address flow from HRA to IMS										
12	Search	Search-10	Search recent history										
13	Search	Search 11	Search on different first name & lastname gives lastname										
14													
15	HRA	HRA-1	Edit patient profile on search page										
16		HRA-2	Verify all HRA data flowed to IMS										
17													

Summary | Bugs | IMS-Search | Interaction | Quick Actions | HRA | Personal Health History ...

	Timestamp	Description	Which Page?	Issue Type	Screenshot	Urgency	Browser	Re-test Date	Testing Comments
2	2/7/13 17:08	Clicking on enter after entering data closes the modal Steps: a) Click on 'Create New Patient' b) Enter patient name and click enter c) Create patient modal is closed	Login	Error or Bug	None	Medium	Firefox	2/21/2013	Pending

Summary | Bugs | IMS-Search | Interaction | Quick Actions | HRA | Personal Health History ...

	Test Scenario Group	Test Case Id	Test Case Description	Test Input	Expected Result	Actual Result	Test Browser	Executed Date	Test Results
2	**Health records**								
3	**Personal Health History**	PH 1	High Cholestrol:	a) Yes and on medication	how old were you at time of dx?	how old were you at time of dx?	Chrome,IE, Firefox		
4				b) Yes but not on medication	how old were you at time of dx?	how old were you at time of dx?	Chrome,IE, Firefox		
5		PH 2	High Blood Pressure:	a) Yes and on medication	what does your blood pressure range, on average?	what does your blood pressure range, on average?	Chrome,IE, Firefox		
6				b) Yes but not on medication	what does your blood pressure range, on average?	what does your blood pressure range, on average?	Chrome,IE, Firefox		
7				a) Yes and on medication	how old were you at time of dx?	how old were you at time of dx?	Chrome,IE, Firefox		
8				b) Yes but not on medication	how old were you at time of dx?	how old were you at time of dx?	Chrome,IE, Firefox		
9		PH 3	Diabetes:	a) Yes and on medication	how old were you at time of dx?		Chrome,Firefox		
10				b) Yes but not on medication	how old were you at time of dx?		Chrome,Firefox		
11				a) Yes and on medication	what type of dm do you have?		Chrome,Firefox		

Assignment No.:- 4

Create Defect Report for Any application or web application

Sample Bug Report

The Sample, Bug/Defect Report given below will give you an exact idea of how to report a Bug in the Bug Tracking Tool?

Here is an Example scenario that caused a Bug:

Let's assume that in your application under test you want to create a new user with user information, for that you need to login into the application and navigate to the USERS menu -> New User, then enter all the details in the 'User form' like, First Name, Last Name, Age, Address, Phone etc.

Once you enter all this information, you need to click on the 'SAVE' button in order to save the user. Now you can see a successful message saying, "New User has been created successfully".

But when you entered into your application by logging in and you have navigated to the USERS menu -> New user, entered all the required information to create the new user and clicked on SAVE button.

BANG! The application crashed and you got one error page on the screen. (Capture this error message window and save it as a Microsoft paint file)

Now, this is a Bug scenario and you would like to report this as a BUG in your Bug-Tracking Tool.

How Will You Report This Bug Effectively?

Sample Bug Report

Here is a sample Bug Report for the above-mentioned example:

(Note that some 'Bug Report' fields might differ depending on your bug tracking system)

SAMPLE BUG REPORT

Bug Name: Application crashes upon clicking the SAVE button while creating a new user.

Bug ID: (It will be automatically created by the BUG Tracking tool once you save this bug).

Area Path: USERS menu -> New Users

Build Number: Version Number 5.0.1

Severity: HIGH (High/Medium/Low) or 1

Priority: HIGH (High/Medium/Low) or 1

Assigned to: Developer-X

Reported By: Your Name

Reported On: Date

Reason: Defect

Status: New/Open/Active (Depends on the Tool you are using)

Environment: Windows 2003/SQL Server 2005

Description: Application crashes upon clicking the SAVE button while creating a new the user, hence unable to create a new user in the application.

Steps to Reproduce:

1) Login into the Application.

2) Navigate to the Users Menu -> New User

3) Filled out all the user information fields.

4) Clicked on the 'Save' button.

5) Seen an error page "ORA1090 Exception: Insert values Error…"

6) See the attached logs for more information (Attach more logs related to the bug..IF any)

7) Also see the attached screenshot of the error page.

Expected Result: On clicking the SAVE button, you should be prompted to a successful message "New User has been created successfully".

(Attach 'application crash' screenshot. IF any)

Save the Defect/Bug in the BUG TRACKING TOOL. You will get a Bug ID that you can use for further bug reference.

Default 'New Bug' mail will go to the respective developer and the default module owner (Team leader or manager) for further action.

How To Write A Good Bug Report? Tips And Tricks

Why a good Bug Report?

If your Bug report is effective, then its chances of getting fixed are higher. So fixing a bug depends upon how effectively you report it. Reporting a bug is nothing but a skill and in this tutorial we will explain how to achieve this skill.

"The point of writing a problem report (bug report) is to get bugs fixed" – By Cem Kaner. If a tester is not reporting a bug correctly, then the programmer will most likely reject this bug stating it as irreproducible. This can hurt the tester's morals and sometimes the ego too. (I suggest not to keep any type of ego. ego's like "I have reported the bug correctly", "I can reproduce it", "Why has he/she rejected the bug?", "It's not my fault" etc.,).

Qualities of a Good Software Bug Report

Anyone can write a Bug report. But not everyone can write an effective Bug report. You should be able to distinguish between an average bug report and a good bug report.

How to distinguish between a good and bad Bug Report? It's very simple, apply the following characteristics and techniques to report a bug.

Characteristics and Techniques

#1) Having a clearly specified Bug Number: Always assign a unique number to each bug report. This, in turn, will help you identify the bug record. If you are using any automated bug-reporting tool then this unique number will be generated automatically each time you report a bug.

Note the number and a brief description of each bug that you reported.

#2) Reproducible: If your bug is not reproducible, then it will never get fixed.

You should clearly mention the steps to reproduce the bug. Do not assume or skip any reproducing steps. The bug which is described Step by step is easy to reproduce and fix.

#3) Be Specific: Do not write an essay about the problem.

Be Specific and to the point. Try to summarize the problem in minimum words yet in an effective way. Do not combine multiple problems even if they seem to be similar. Write different reports for each problem.

Effective Bug Reporting

Bug reporting is an important aspect of Software Testing. Effective Bug reports communicate well with the development team to avoid confusion or miscommunication.

Good Bug report should be **clear and concise** without any missing key points. Any lack of clarity leads to misunderstanding and slows down the development process as well. Defect writing and reporting is one of the most important but neglected areas in the testing life cycle.

Good writing is very important for filing a bug. The most important point that a tester should keep in mind is **not to use a commanding tone** in the report. This breaks morale and creates an unhealthy work relationship. Use a suggestive tone.

Don't assume that the developer has made a mistake and hence you can use harsh words. Before reporting, it is equally important to check if the same bug has been reported or not.

A duplicate bug is a burden in the testing cycle. Check out the whole list of known bugs. At times, the developers may be aware of the issue and ignore it for future releases. Tools like Bugzilla, which automatically searches for duplicate bugs, can also be used. However, it is best to manually search for any duplicate bug.

The import information that a bug report must communicate is **"How?" and "Where?"** The report should clearly answer exactly how the test was performed and where the defect occurred. The reader should easily reproduce the bug and find out where the bug is.

Keep in mind that the **objective of writing a Bug report** is to enable the developer to visualize the problem. He/She should clearly understand the defect from the Bug report. Remember to provide all the relevant information that the developer is seeking.

Also, bear in mind that a bug report would be preserved for future use and should be well written with the required information. **Use meaningful sentences and simple words** to describe your bugs. Don't use confusing statements that waste the time of the reviewer.

Report each bug as a separate issue. In case of multiple issues in a single Bug report, you can't close it unless all the issues are resolved.

Hence, it is best to **split the issues into separate bugs**. This ensures that each bug can be handled separately. A well-written bug report helps a developer to reproduce the bug at their terminal. This will help them diagnose the issue as well.

How To Report A Bug?

Use the following simple Bug report template:

This is a simple Bug report format. It may vary depending upon the Bug report tool that you are using. If you are writing a bug report manually then some fields need to be mentioned specifically like the Bug number – which should be assigned manually.

Reporter: Your name and email address.

Product: In which product you found this bug.

Version: The product version, if any.

Component: These are the major sub-modules of the product.

Platform: Mention the hardware platform where you found this bug. The various platforms like 'PC', 'MAC', 'HP', 'Sun' etc.

Operating system: Mention all the operating systems where you found the bug. Operating systems like Windows, Linux, Unix, SunOS, and Mac OS. Also, mention the different OS versions like Windows NT, Windows 2000, Windows XP etc, if applicable.

Priority: When should a bug be fixed? Priority is generally set from P1 to P5. P1 as "fix the bug with the highest priority" and P5 as " Fix when time permits".

Severity: This describes the impact of the bug.

Types of Severity:

- **Blocker:** No further testing work can be done.
- **Critical:** Application crash, Loss of data.
- **Major:** Major loss of function.
- **Minor:** Minor loss of function.
- **Trivial:** Some UI enhancements.
- **Enhancement:** Request for a new feature or some enhancement in the existing one.

Status: When you are logging the bug into any bug tracking system then by default the bug status will be 'New'.

Later on, the bug goes through various stages like Fixed, Verified, Reopen, Won't Fix, etc.

Assign To: If you know which developer is responsible for that particular module in which the bug occurred, then you can specify the email address of that developer. Else keep it blank as this will assign the bug to the module owner, if not the Manager will assign the bug to the developer. Possibly add the manager's email address to the CC list.

URL: The page URL on which the bug occurred.

Summary: A brief summary of the bug, mostly within 60 words or below. Make sure your summary is reflecting on what the problem is and where it is.

Description: A detailed description of the bug.

Use the following fields for the description field:

- **Reproduce steps:** Clearly, mention the steps to reproduce the bug.
- **Expected result:** How the application should behave on the above-mentioned steps.
- **Actual result:** What is the actual result of running the above steps i.e. the bug behavior.

These are the important steps in the bug report. You can also add "Report Type" as one more field which will describe the bug type.

Report Types include:

1) Coding error
2) Design error
3) New Suggestion
4) Documentation issue
5) Hardware problem

Important Features in Your Bug Report

Given below are the important features in the Bug report:

#1) Bug Number/id

A Bug number or an identification number (like swb001) makes bug reporting and the process of referring to bugs much easier. The developer can easily check if a particular bug has been fixed or not. It makes the whole testing and retesting process smoother and easier.

#2) Bug Title

Bug titles are read more often than any other part of the bug report. This should explain all about what comes in the bug. The Bug title should be suggestive enough that the reader can understand it. A clear bug title makes it easy to understand and the reader can know if the bug has been reported earlier or has been fixed.

#3) Priority

Based on the severity of the bug, a priority can be set for it. A bug can be a Blocker, Critical, Major, Minor, Trivial, or a suggestion. Bug priorities can be given from P1 to P5 so that the important ones are viewed first.

#4) Platform/Environment

OS and browser configuration is necessary for a clear bug report. It is the best way to communicate how the bug can be reproduced.

Without the exact platform or environment, the application may behave differently and the bug at the tester's end may not replicate on the developer's end. So it is best to clearly mention the environment in which the bug was detected.

#5) Description

Bug description helps the developer to understand the bug. It describes the problem encountered. A poor description will create confusion and waste the time of the developers as well as testers.

It is necessary to communicate clearly about the effect of the description. It's always helpful to use complete sentences. It is a good practice to describe each problem separately instead of crumbling them altogether. Don't use terms like "I think" or "I believe".

#6) Steps to Reproduce

A good Bug report should clearly mention the steps to reproduce. These steps should include actions that may cause the bug. Don't make generic statements. Be specific on the steps to follow.

A good example of a well-written procedure is given below

Steps:

- Select product Abc01.
- Click on Add to cart.
- Click Remove to remove the product from the cart.

#7) Expected and Actual Result

A Bug description is incomplete without the Expected and Actual results. It is necessary to outline what the outcome of the test is and what the user should expect. The reader should know what the correct outcome of the test is. Clearly, mention what happened during the test and what the outcome was.

#8) Screenshot

A picture is worth a thousand words. Take a Screenshot of the instance of failure with proper captioning to highlight the defect. Highlight unexpected error messages with light red color. This draws attention to the required area.

Some Bonus Tips To Write A Good Bug Report

Given below are some additional tips on how to write a good Bug report:

#1) Report the problem immediately

If you find any bugs while testing, then you do not need to wait to write a detailed bug report later. Instead, write a bug report immediately. This will ensure a good and reproducible Bug report. If you decide to write the Bug report later on then there is a higher chance to miss the important steps in your report.

#2) Reproduce the bug three times before writing a Bug report

Your bug should be reproducible. Make sure that your steps are robust enough to reproduce the bug without any ambiguity. If your bug is not reproducible every time, then you can still file a bug mentioning the periodic nature of the bug.

#3) Test the same bug occurrence on other similar modules

Sometimes the developer uses the same code for different similar modules. So there is a higher chance for the bug in one module to occur in other similar modules as well. You can even try to find the more severe version of the bug you found.

#4) Write a good bug summary

Bug summary will help the developers to quickly analyze the bug's nature. A poor quality report will unnecessarily increase development and testing time. Communicate well with your bug report summary. Keep in mind that the bug summary can be used as a reference to search for the bug in the bug inventory.

#5) Read the Bug report before hitting the Submit button

Read all the sentences, wordings and steps that are used in the bug report. See if any sentence is creating ambiguity that can lead to misinterpretation. Misleading words or sentences should be avoided in order to have a clear bug report.

#6) Do not use abusive language.

It's nice that you did a good work and found a bug but do not use this credit for criticizing the developer or to attack any individual.

Sample Bug Reports For Web And Product Applications

Bug report sample 1: Web Project bug report

Summary: In CTR (Click through ratio) 'Total' row calculation is wrong

Product: Example product

Version: 1.0

Platform: PC

URL: (Provide url of page where bug occurs)

OS/Version: Windows 2000

Status: NEW

Severity: Major

Priority: P1

Component: Publisher stats

Assigned To: developer@example.com

Reported By: tester@example.com

CC: manager@example.com

Bug Description:

Reproduce steps:

1) Go to page: (Provide URL of page where bug occurs)

2) Click on 'Publisher stats' link to view publisher's revenue detail stats date wise.

3) On page (Provide URL of page where bug occurs) check CTR value in 'Total' row of CTR stats table.

Actual result: Calculation of 'Total' row in CTR table is wrong. Also Individual row CTR for each publisher is not truncated to 2 digits after decimal point. It's showing CTR like 0.042556767

Expected result: Total CTR= (Total clicks/Total searches)*100

[Attach bug screenshot if any]

Please fix the bug.

****

Sample bug report 2: Application product Bug report sample

Application testing scenario:

Lets assume in your application you want to create a new user with his/her information, for that you need to logon into the application and navigate to USERS menu > New User, then enter all the details in the User form like, First Name, Last Name, Age, Address, Phone etc. Once you enter all these need to click on SAVE button in order to save the user and you can see a success message saying, "New User has been created successfully". Now you entered into your application by logging in and navigate to USERS menu > New user, entered all the information and clicked on SAVE button and now the application crashed and you can see one error page on the screen, now you would like to report this BUG.

Now here is how we can report bug for above scenario:

Bug Name: Application crash on clicking the SAVE button while creating a new user.

Bug ID: The BUG Tracking tool will automatically create it once you save this.

Area Path: USERS menu > New Users

Build Number:/Version Number 5.0.1

Severity: HIGH (High/Medium/Low)

Priority: HIGH (High/Medium/Low)

Assigned to: Developer-X

Created By: Your Name

Created On: Date

Reason: Defect

Status: New/Open/Active – Depends on the Tool you are using

Environment: Windows 2003/SQL Server 2005

Description:

Application crash on clicking the SAVE button while creating a new user, hence unable to create a new user in the application.

Steps To Reproduce:

1) Logon into the application
2) Navigate to the Users Menu > New User
3) Filled all the fields
4) Clicked on ‘Save’ button
5) Seen an error page “ORA1090 Exception: Insert values Error…”
6) See the attached logs for more information
7) And also see the attached screenshot of the error page.

Expected: On clicking SAVE button should be prompted to a success message “New User has been created successfully”.

Save the defect/bug in the BUG TRACKING TOOL.

Assignment No.:- 5

Installation of Selenium grid and selenium Web driver java eclipse (automation tools).

Managing user experience is pivotal for software development. Test automation enables user preferences and convenience to remain at the center of the development process while saving time and effort. That is why comprehensive automation testing has become necessary to retain customers and meet their expectations. With significantly shorter time frames for development, Selenium Testing, in particular, has become an integral part of the development to facilitate automated testing of web applications.

Selenium is the most popular automated tool in existence today. 59.5% of people consider Selenium for Cross Browser Testing because of the robustness and flexibility it offers by supporting multiple languages like Java, C#, Python, Perl, Ruby, etc. However, a majority (67%) of the Selenium users prefer Java as their language for Selenium Testing.

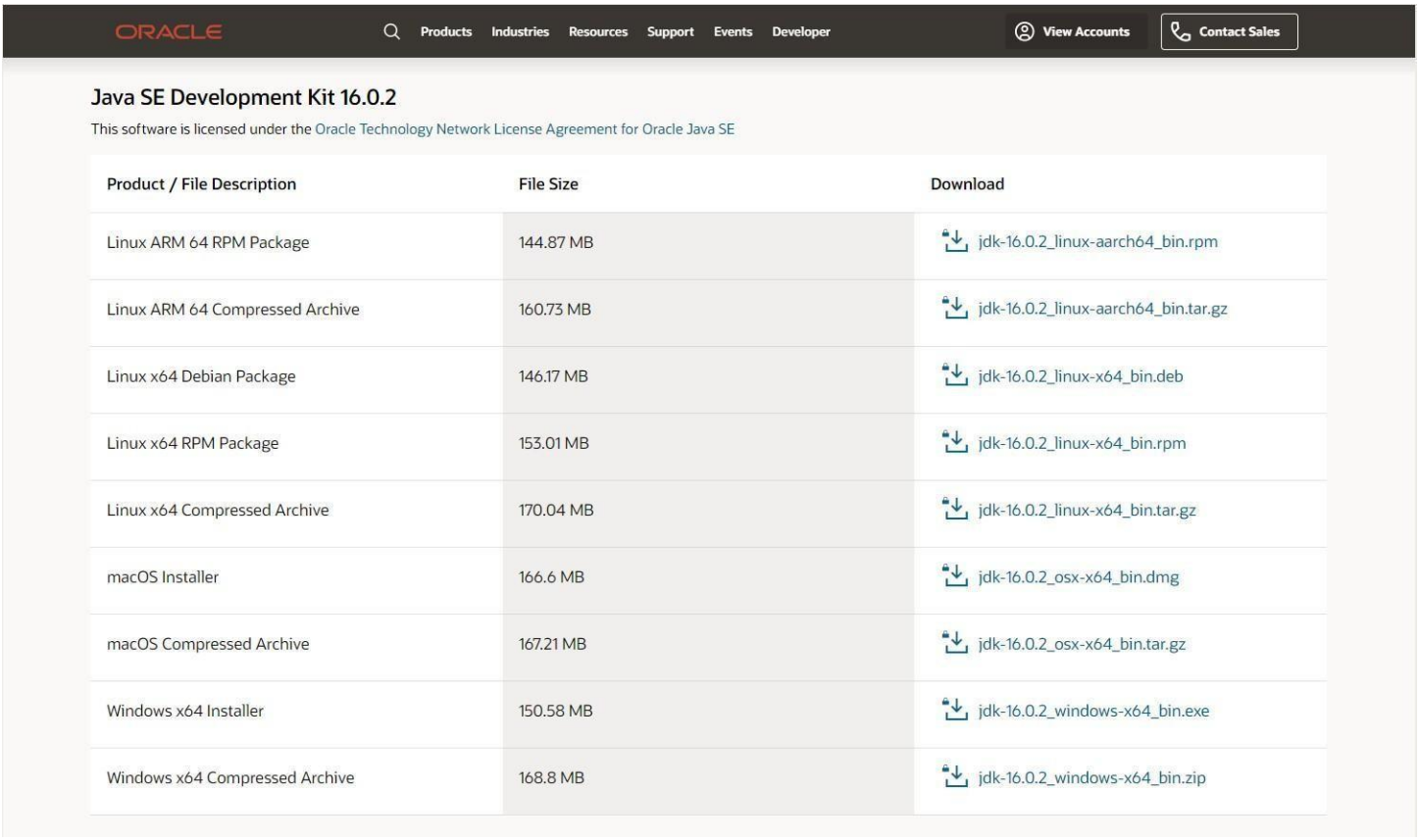
This article discusses how to configure Selenium in Eclipse to use Selenium for Java.

Table of Contents

- Prerequisites for configuring Selenium in Eclipse
- How to configure Selenium in Eclipse
- Conclusion

Prerequisites for configuring Selenium in Eclipse

Install JavaDownload Java SE Development Kit 16.0.2 according to the Windows, Linux, or macOS platform being used.

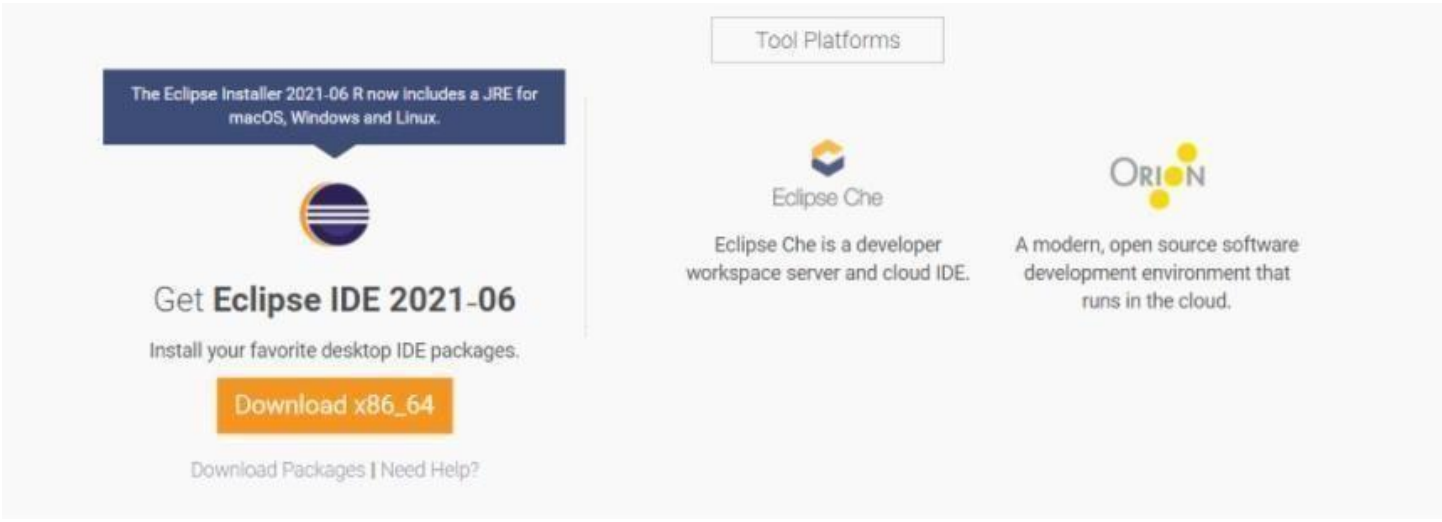


Source

Run the JDK Installer by double-clicking on the file name in the download location and following the instructions on the instruction wizard. Alternatively, silently install JDK by entering the following command:

jdk.exe /s

Install Eclipse IDE



Install Selenium

Download and Install Selenium to be set up in Eclipse.

Install Browser Driver

For Cross Browser Testing, download the relevant Browser Driver – ChromeDriver (for Chrome), GeckoDriver (for Firefox), SafariDriver(for Safari), and InternetExplorerDriver and MSEdgeDriver (IE and Edge respectively). Place these Browser Driver files in a directory that is part of the environment PATH. This will allow a command-line call to the programs to execute them irrespective of the working directory.

Install Java Language Bindings

Version 3.141.59 (2018)

Changelog

API Docs



How to configure Selenium in Eclipse

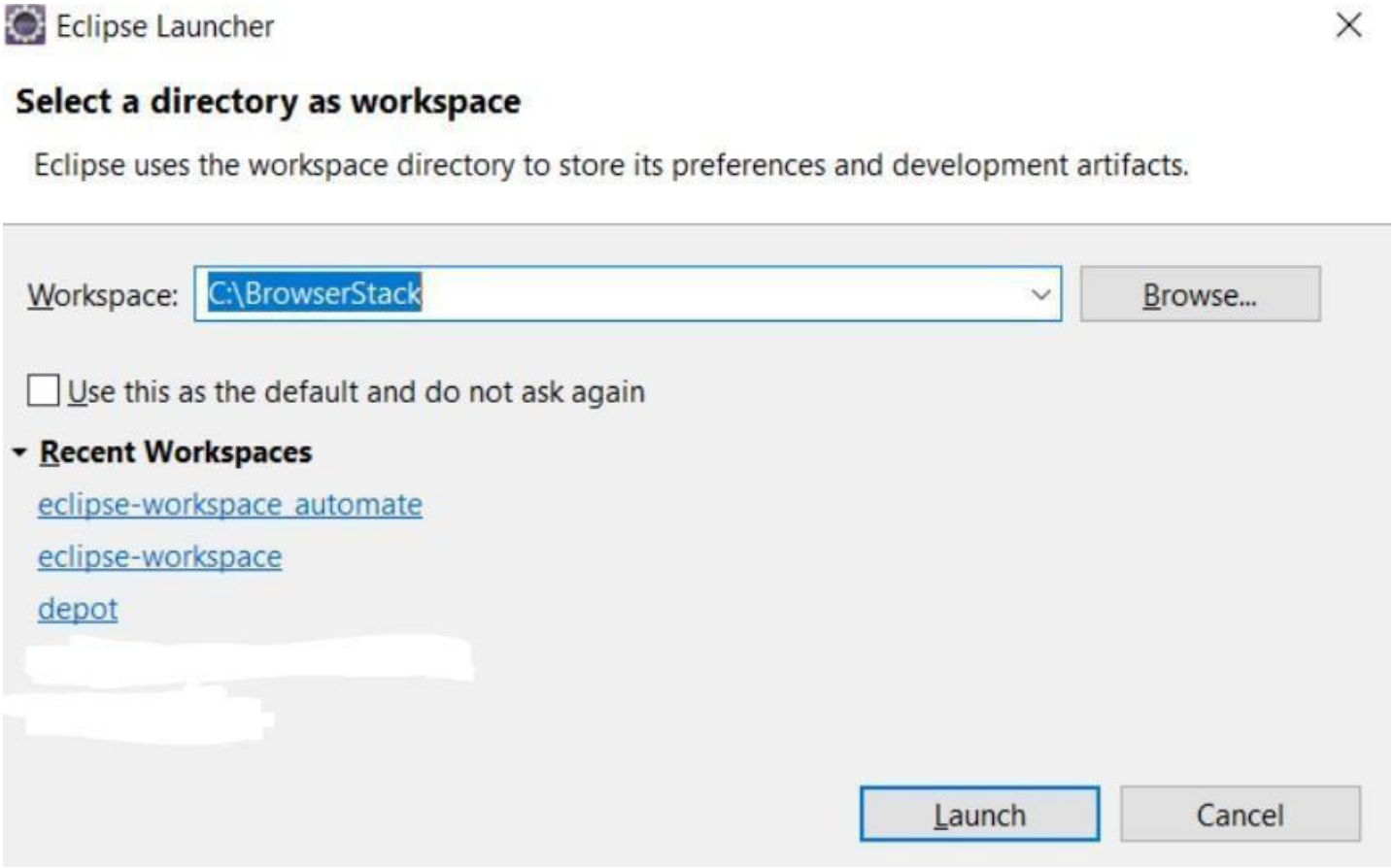
Here are the steps to configure Selenium Webdriver with Eclipse:

Step 1: Launch Eclipse

To launch Eclipse double click on the eclipse.exe file in the download location.

Step 2: Create Workspace in Eclipse

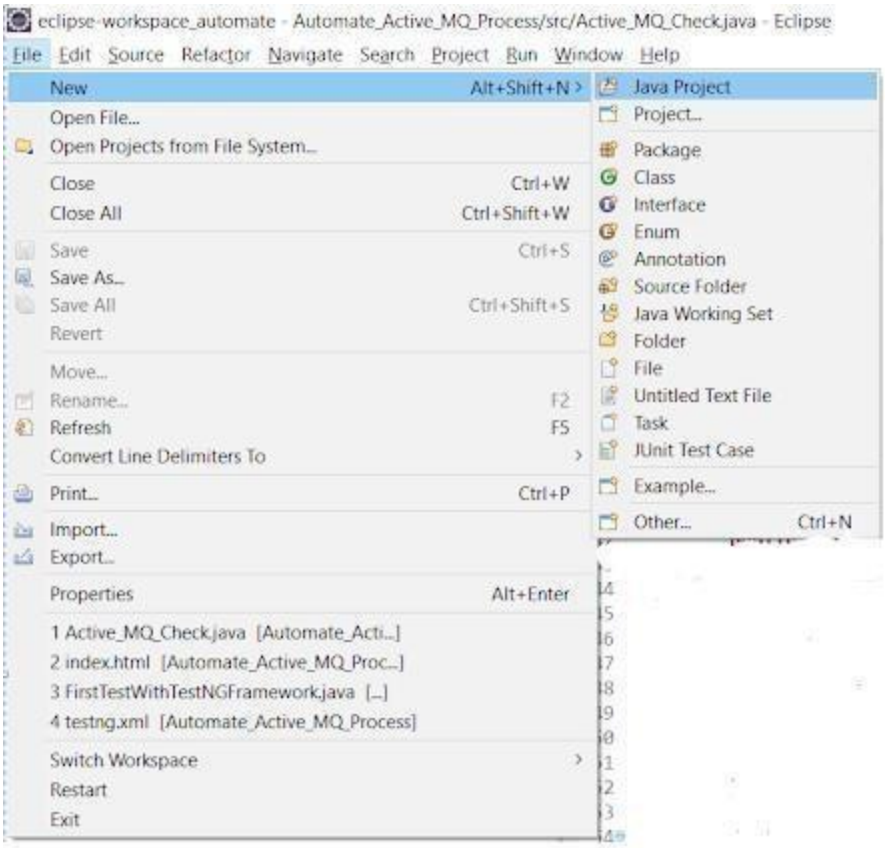
This workspace named “C:\BrowserStack” is like any other folder, which will store all the test scripts. Launch the BrowserStack workspace.



Creating Workspace in Eclipse

Step 3: Create New Java Project in the BrowserStack Workspace

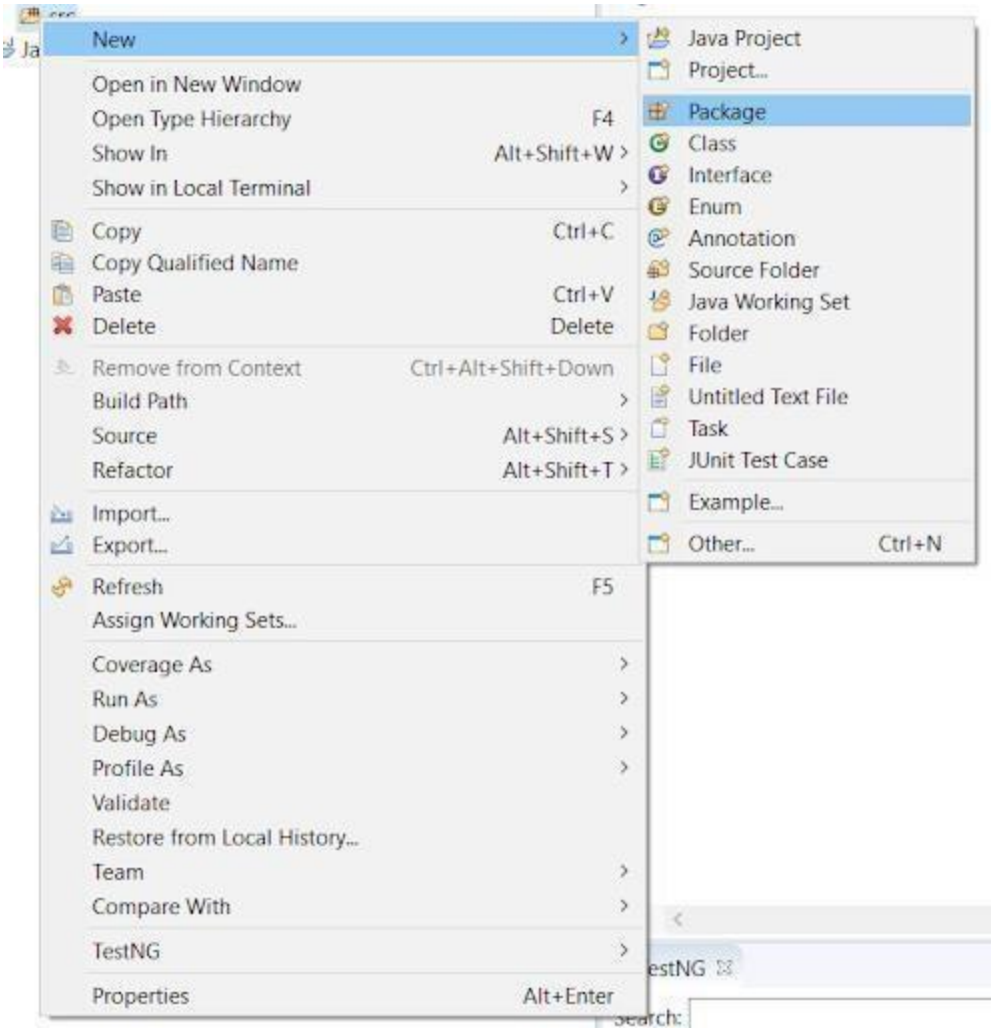
Create a new Java Project by clicking on File > New > Java Project and name it.



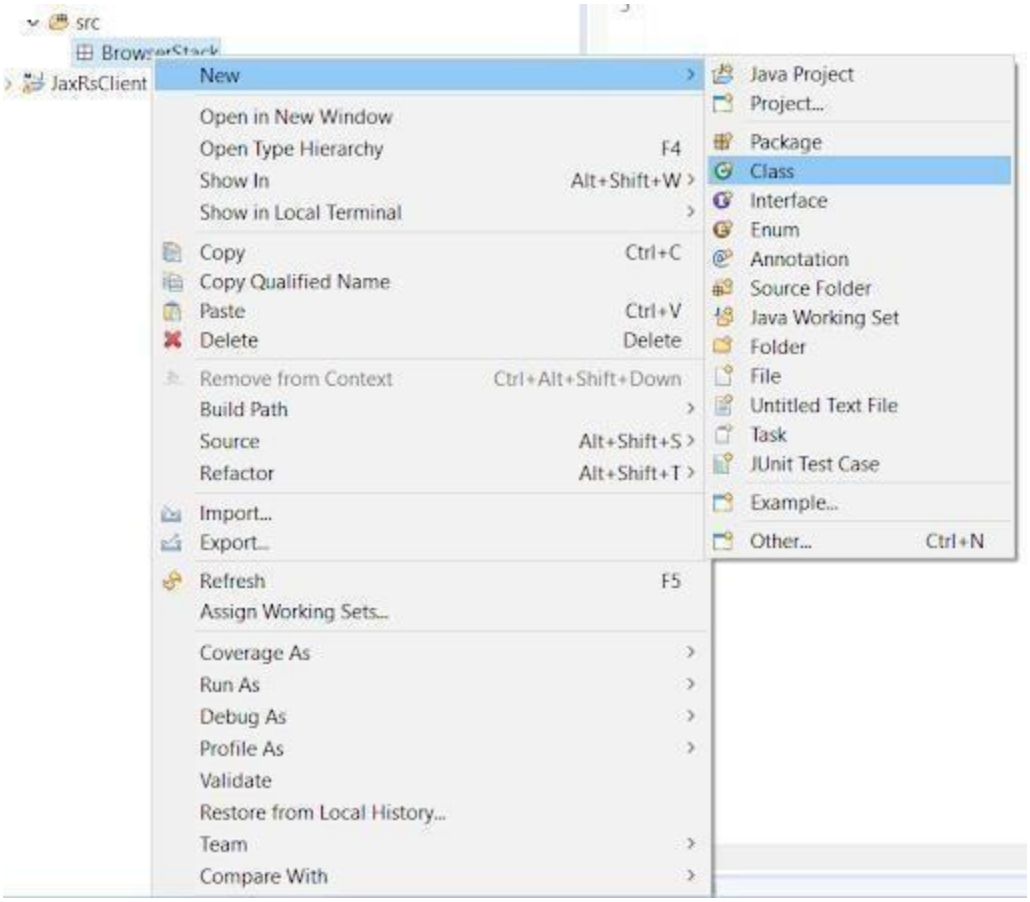
Creating a new Java Project

Step 4: Create Package and Class under the Java Project

By clicking on the src folder (which is the source folder), create a new package and name it (BrowserStack). Then right-click on the package name and create a class.



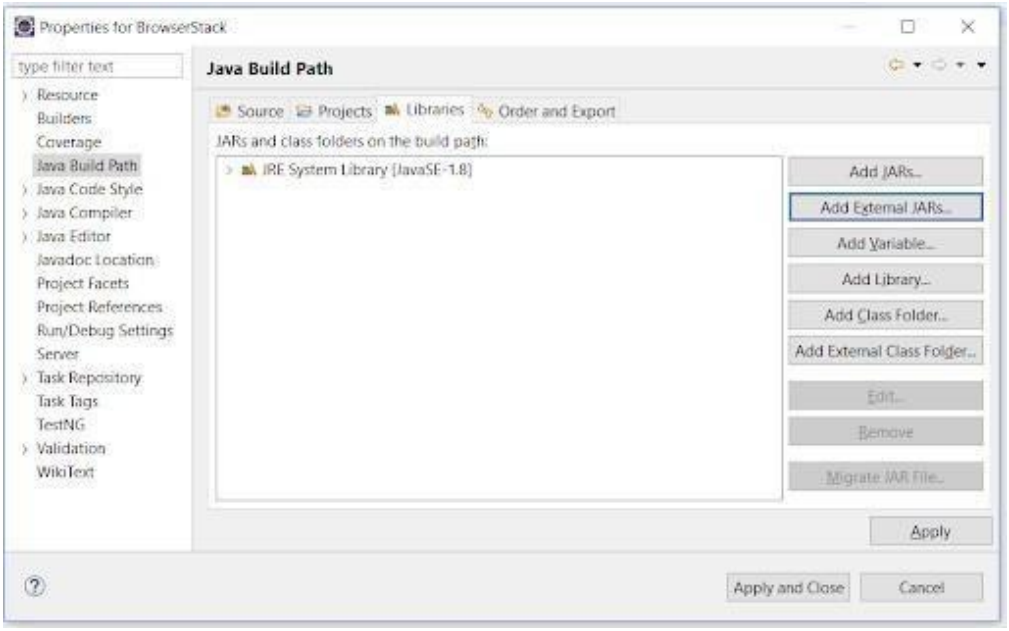
Creating Package in the Java Project



Creating Class in the BrowserStack Package

Step 5: Add Selenium JARs to the Java Project in Eclipse

To add the Selenium Jars to the BrowserStack Java right click on the BrowserStack Project folder and select the Properties option. In the properties window, click on the Java Build Path and Add External JARs. Browse and add the downloaded Selenium JARs i.e. Client Combined JAR and all the JARs under the Libs folder, then click Apply and Close.



Adding Selenium JARs in the BrowserStack Project

This configures Selenium with Eclipse, making it ready to execute the first test script.

Conclusion

Automation testing using Selenium with Java has simplified software development. Being an open-source tool, it provides an opportunity to speed up the time of execution and remove errors for a better user experience.

Java is quite popular among the developers, given the extensive support being available, making it a preferred choice as a Selenium Client Language Binding. Selenium with Java has proved helpful in optimizing regression testing and cross browser testing. With the support of Continuous Integration tools like Jenkins, Selenium with Java can be used in continuous delivery models.

However, to make testing more efficient and get accurate results every time, all Selenium tests must be run on real browsers and devices for accurate results. Start running tests on 2000+ real browsers and devices on BrowserStack's real device cloud. Run parallel tests on its Cloud Selenium Grid to get faster results without compromising on accuracy. Identify all bugs and offer a high-end UX/UI to the users by running automated tests in real user conditions with BrowserStack Automate.